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EchoMap

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Abstract

The Noise Pollution Mapping and Analysis System is a system that is developed for the purpose of measuring and visualizing noise levels in different areas through the use of data collected via smartphones or sensors. The system logs the sound levels along with the time and location, trains an algorithm to tell the type of noise, and finally presents the data on a digital map that the user can interact with. Besides, a time-lapse feature is also included to visualize the noise level changes during a day. The current project is directed towards assisting people as well as city planners in comprehending noise dispersal and trends in the metropolitan areas. The system not only leads to improved decision-making regarding the reduction of noise pollution but also plays a role in making cities more sustainable and habitable.

Executive Summary

The Noise Pollution Mapping and Analysis System is a project designed to collect, analyze, and visualize environmental noise data using smartphones or external sensors. The main purpose of this project is to help users, researchers, and local authorities understand noise levels in different areas and take necessary steps to reduce noise pollution. The system records sound intensity in decibels (dB), captures the time and location of the noise, and uses machine learning to classify the type of noise source such as traffic, people, or construction.

The mechanism of the system is such that it gathers the current sound data from the consumers via a mobile or web interface. Every sound measurement is kept in a cloud database with the associated GPS coordinates and the time it was taken. The data is clean of any unwanted noise and then the trained machine learning model analyzes it putting it into specific categories. The outcome is shown on an interactive map in the form of color-coded markers or heatmaps depicting the quietest or the noisiest areas.

The time-lapse visualization is another important aspect of the system which enables the users to see the noise levels in a certain area changing with time. The identification of noise patterns, like the rush hours for traffic or the quiet times of the day, is facilitated by this feature. In addition, the system possesses a number of charts and statistics that give a clear picture of average noise level and noise type distribution in the chosen places.

The main goal of the project is to create a solution that is not only lightweight but also accessible for everyone who owns a smartphone. Unlike the conventional noise monitoring systems, which are all dependent on expensive sensors at fixed locations, this new system is more flexible and scalable as it collects data from the users. The project considerably focuses on the use of open-source technologies and cloud storage to minimize the costs and at the same time, to allow easy integration with other systems in the future.

The initiative is aligned with the United Nations Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities aimed at making urban areas more inclusive, secure, and resilient. The system, through the mapping of noise data, assists planners and policymakers in making informed decisions regarding zoning, traffic control, and environmental regulations.

The anticipated result of the project would be an application that works on the web and mobile platforms, and that is able to record, analyze, and show the noise data live. The project will also provide documentation covering the design, implementation, and testing phases, as well as a trained model for the classification of noise. In the future, this system could be turned into a commercial or research plat-

form that offers citywide noise analytics in real-time, noise modeling, and the integration of IoT devices. It will also help educational institutes and environmental groups to carry out research on noise pollution and its impacts.

This urban noise pollution monitoring project makes use of data collection, machine learning, and visualization to develop an efficient and easy-to-use system. It not only gives important information to both individuals and authorities but also plays a role in enhancing the livability of noisy areas through its implementation.

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Chapter 1 Introduction

In contemporary metropolitan areas, one of the most irritating and neglected environmental issues has to be noise pollution. The human being has to face the bitter end of the spectrum when he/she is exposed to noise of high decibels for a long time. Such an exposure can cause stress, sleep disturbances, hearing loss, and decreased efficiency, all physical and mental health problems. But still, noise monitoring technology is not that effective and there are only a few cities that can boast of having any real-time noise data collection and visualization tools. EchoMap is the response to all these issues by offering a system for noise mapping and analysis that is interactive and shows the environmental sound levels in various geographical regions. It provides an easy-to-use web interface for individuals, researchers, and government bodies to collect, analyze, and read environmental noise data. With the use of geospatial visualization and analytics, EchoMap gives users the power to spot noise hotspots, monitor patterns, and take educated actions to make urban soundscapes healthier.

1.1 Purpose of this Document

The document is meant to give a detailed summary about the EchoMap system, presenting its reasons, targets, and the technical method in its development as well. It is a thorough map that sets the problem to be solved, the boundaries of the solution, and the process to be followed in order to reach the project's objectives. Thus, the report is a guide for the system's architecture, design, and implementation that makes it possible for every phase of development to be in line with the set objectives. Also, it reveals the reasoning behind the main design and processing techniques, and the tools used. The document not only plays the role of a technical reference but also points out the larger environmental and social significance of EchoMap, thus asserting its function of helping make urban noise management and public health awareness decisions based on reliable information.

1.2 Intended Audience

This paper is directed towards a wide range of readers. Among others, the document will be read by both technical and non-technical stakeholders who are involved or affected by the EchoMap project. The main audience consists of the project supervisors, evaluators, and academic reviewers who will judge the system in terms of urban informatics design, implementation as well as contribution to the environmental field. Likewise, the document will be useful for software developers and engineers who might refer to it to know more about the system's architecture, data flow, and implementation details for further enhancement or duplication. Researchers in the environmental field, policymakers, and urban planners

are a few representatives of the secondary audience who might benefit from the information and results that the EchoMapwill provide, as they are likely to be data-driven decision-makers and planners. Finally, the document may also act as an educational tool for students and upcoming developers who want to work on similar projects applying data science, web technologies, and GIS to real-world environmental problems.

1.3 Definitions, Acronyms, and Abbreviations

List all important definitions, acronyms, and abbreviations used in this document.

API: Application Programming Interface

DB: Database

db: Decibel-Unit used tomeasure intensity of sound

GIS: Geographic Information system

GPS: Global positioning system

UI: User interface

UX: User experience

ML: Machine learning

IoT: Internet of things

CRUD: Create, Read, Update, Delete

HTTP: Hypertext Transfer Protocol

GUI: GRaphical user interface

SDK: Software development kit

1.4 Conclusion

To sum up, this chapter has laid down the groundwork for the future development of the EchoMap project, which was mainly based on the motivation, purpose, and the direction of the system that it intended to take. With interactive geospatial visualization at its core, the chapter has argued for a modern, data-driven method to be used in monitoring and managing urban noise pollution. The discussion has also created the foundations for the objectives, scope, and importance of EchoMap, pointing out its possible role in raising awareness about the environment and in urban planning. Moreover, the document's purpose and target audience were identified so that technical development could be in the line with the stakeholders' expectations. The concepts and definitions provided in this chapter are like a map for the next chapters, which will cover more about the literature background, design methodology, implementation, and evaluation processes of the system.

Chapter 2 Project Vision

2.1 Problem Domain Overview

The Noise Pollution Mapping and Analysis System is an application that has the capability to document, scrutinize, and make visible the levels of environmental noise in various places.Each measurement consists of the noise level in decibels, the geographical location, and the time.The system then uploads the data to the cloud database for storage. A machine learning model is employed to discern different types of noises, for example, traffic, people talking, or construction. The data is subsequently presented on a map as a heatmap, where the users can visualize the noise levels in different areas. Additionally, a time-lapse feature is integrated into the system through which the users can observe the variation of noise levels at different times of the day.This system aims at enabling the users, city planners, and researchers to see and comprehend the noise levels in their environment such that the necessary steps are taken towards noise pollution reduction.

2.2 Problem Statement

Cities around the world are suffering from noise pollution problems which for the most part, are the result of high environmental noise levels. People who live in areas exposed to such noise experience health and quality of life impacts. The existing noise monitoring systems are not able to cope with this situation as their operation relies on costly fixed sensors that only monitor small spots. Monitoring real-time noise levels and patterns in the area has no straightforward approach for the public or the local governments.The goal of this project is to overcome the aforementioned challenge by creating a system that collects the noise data and presents it on a map

2.3 Problem Elaboration

Noise pollution affects human health, comfort, and productivity. Sources of noise include traffic, construction, industries, and loud gatherings. Monitoring these sources is important, but traditional systems require fixed sensors, which are costly and limited in coverage. The project will work on several subproblems like Data collection (Getting accurate noise readings from microphones or sensors with their location and time.),Noise classification (Using machine learning to identify what kind of sound was recorded.Data visualization: Showing the noise data clearly on a map using colors to represent intensity.),Time analysis (Allowing users to see how noise levels change during the day through a time-lapse feature.),Data management (Storing and retrieving data efficiently from a cloud database.).By solving

these sub-problems, the system will make it possible to record, analyze, and visualize noise in a simple and accessible way.

2.4 Goals and Objectives

2.4.1 Goals

- To create an advanced system that would be able to gather, categorize, and show environmental noise data instantly.
- To provide an interactive noise map where both the public and urban planners can access the noise levels.
- To analyze and draw conclusions from the noise patterns over time for better public awareness and understanding.

2.4.2 Objectives

- To gather the noises by location and time using either the smartphones or the sensors.
- To use machine learning to process and categorize the recorded sounds.
- To keep entire data taken in a cloud database.
- To depict noise levels through a heatmap.
- To make a time-lapse feature in which the day's noise would be shown with the changes.
- To develop a user-friendly web interface that would show the progress of the work and the summary of the results.

2.5 Project Scope

The project aims to create a system, which is based on the web and mobile devices, to pinpoint and analyze the noise pollution in a certain region. The functionalities of the system will be as follows:

- we will use existing noise data as well as user provided data
- Location and time will be attached to each noise entry.
- A machine learning model will be used to identify the noise category.
- An interactive heatmap will show the different noise levels.
- By using a time-lapse feature, one will be able to see the historical silence data.

- Users will receive concise and visual reports.

The project will be utilizing a web stack consisting of HTML, CSS, JavaScript, and either React.js or Flask. Python packages like Librosa and scikit-learn will be employed for the tasks of sound categorization and processing. A database in the cloud, like Firebase or MongoDB, will be used for data storage, and then, data will be visualized through Leaflet.js or Mapbox.The outputs of the project will comprise:

- A fully functional web app that is capable of gathering and showing noise data.
- A machine learning model that has been trained for noise type classification.
- A report of documentation that describes design, implementation, and testingA report of documentation that describes design, implementation, and testing
- A brief user guide describing the usage of the system.

The project scope explicitly states the following as out of the scope:

- Connection with the infrastructure of government monitoring systems.
- The project will not cover the aspect of large-scale noise monitoring in a city area.
- There will be no development of mobile apps for iOS or Android platforms.
- Alerts or notifications that are real-time will be considered as future work

This project scope will be used as a guide throughout the development

2.6 Sustainable Development Goal (SDG)

This initiative aligns with the United Nations Sustainable Development Goal 11: Sustainable Cities and Communities. The system is a tool for municipalities to oversee and control eco-noise, which is a requirement for the formation of sound and sustainable habitats. The system's collection and distribution of noise data creates a society that is noise aware and takes part in the making of less noisy and healthier cities through planning.

Figure 2.1: Our priority focus is on Sustainable Development

2.7 Constraints

There are some limitations regarding technology and finance alongside operations that might impact the project's development and performance. The following areas are the limitations mentioned:

- Noise measuring precision relies heavily on the kind of microphone that each smartphone or sensor has.
- The accuracy of noise categorization may be adversely impacted by background noise.
- Unoptimized processing of huge datasets might result in such slow system performance.
- The accuracy of GPS may differ according to signal strength and device capabilities.
- The use of cloud services' free versions (for example Firebase or Mapbox) may be restricted in terms of usage
- Large-scale tests or deployment cannot be funded
- The machine learning model requires a large amount of labeled data to achieve a good level of accuracy, and collecting such data may take a long time.

2.8 Business Opportunity

Noise pollution is recognized as an increasing issue in metropolitan areas and there are scarcely any practical ways for monitoring it. This project would encourage the emergence of a low-cost and community-based noise monitoring system. In the long run, the platform can be developed into a fully-fledged service with subscription plans and data dashboards for businesses. Also, it might partner with noise-resistant IoT sensors and provide real-time alerts for the detection of breaches in noise regulations.Through the provision of both precise and graphical noise data, the system is likely to facilitate more effective urban management, raise the public's environmental consciousness, and create new avenues in the environmental technology sector.

2.9 Conclusion

The Noise Pollution Mapping and Analysis System is a platform that allows the integration of web and mobile technologies for the collection, classification, and visualization of noise data from different locations. Its main feature is to provide people with an easy and inexpensive method to get an idea of noise level variations depending on location and time.The above-discussed chapter gave an outline of the project vision, the problem it tries to solve, and its goals, objectives, and scope. Likewise, it

mentioned the constraints, both technical and operational, which might impact the system and discussed the business prospects which could be realized through the adoption of such a solution.The system makes the city development plan more eco-friendly by raising awareness of environmental noise and allowing agencies to devise better living areas. The project merges data gathering, machine learning, and visualization, thus providing a foundation for further studies and practical uses in environmental monitoring.

Chapter 3 Literature Review / Related Work

3.1 Definitions, Acronyms, and Abbreviations

List all important definitions, acronyms, and abbreviations used in this document.

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GPS: Global positioning system

UI: User interface

UX: User experience

ML: Machine learning

IoT: Internet of things

CRUD: Create, Read, Update, Delete

HTTP: Hypertext Transfer Protocol

GUI: GRaphical user interface

SDK: Software development kit

3.2 Detailed Literature Review

3.2.1 Noise pollution effect on children

3.2.1.1 Summary

The review by Gupta et al.[1] titled "Noise Pollution and Impact on Children Health" (2018) studies the acoustical situation in the ambiance and its influence on people's health during their childhood, particularly the newborns, infants, and school-going kids. It gives a thorough account of both the auditory effects (e.g. noise-induced hearing loss) and the non-auditory ones (such as sleep disturbance, cognitive impairment, cardiovascular stress, and adverse birth outcomes). Urban noise pollution sources are still in the center of voices from the authors, and these include traffic, industrial machinery, loudspeakers, etc. They also suggest interventions like zoning, soundproofing, ear protection, and public education, besides calling for stricter noise regulation and monitoring.

3.2.1.2 Critical Analysis

The article provides a robust synthesis of the existing literature to highlight the dangers that noise exposure brings especially to the most sensitive groups such as children. Besides, the main advantage is the article's extensive coverage of noise harm effects from the physical, psychological, developmental, and social aspects. In contrast, the review, being a collection of previous studies, is at the same time limited by the quality and heterogeneity of the findings it cites. Sometimes, the authors make strong policy recommendations without systematically ranking the evidence or addressing the feasibility of implementation in low-resource settings. Moreover, while they mention that "areas of noise pollution must be identified," the review does not go into detail about technical or methodological difficulties concerning noise mapping, spatial modeling, or how to incorporate local factors (urban morphology, topography, cultural practices) into assessments.

3.2.1.3 Relationship to the proposed work

The article endorses the necessity and significance of your study: it explicates that the revelations of spatial "hotspots" of noise exposure are a fundamental action before any suppression measures. The mapping activity can transform their demand to "detect areas of noise pollution" into a practical task with the aid of spatially resolved data. In addition, by connecting your mapped data to health risk thresholds (for children, pregnant women, etc.), we can make the review's theoretical recommendations into actionable interventions (e.g. zoning, targeted noise barriers, community awareness in high-risk zones) with your help. Local mapping can also correct one of the review's deficiencies by demonstrating how noise changes in real urban morphology and thus, providing empirical grounding for regulation in your area.

3.2.2 Noise pollution and urban planning

3.2.2.1 Summary

The authors of the article[2] state that urban planning and design have a strong impact on the location and timing of noise pollution in cities. The origin of this interaction lies in urban density, street layout, land use patterns, building thickness, greenery, and traffic. In addition, the authors discuss the various measurement and sampling strategies for noise in the urban context while pointing out that adequate merging of noise data with urban features can enhance prediction, analysis, and noise exposure prevention. Finally, they propose the combination of urban planning and acoustic standards as a road to the establishment of noise-free and environmentally-friendly cities.

3.2.2.2 Critical Analysis

This paper is important since it connects and combines the two areas—urban planning and acoustics—which means it will be easier to look at the problem from a broad angle regarding how city design influences noise exposure. Among other things, its analysis of sampling methods is a great merit, as it makes the pros and cons very clear (spatial resolution, measurement effort, and micro-environmental heterogeneity). Still, the paper has its downside through being mostly theoretical: in this regard, it lists case studies but does not properly deal with empirical backing of the suggested relationships (i.e. how far planning measures can actually reduce noise in practice). Furthermore, the review insists on top-level design advice but does not really talk about the difficulties of implementation in under-resourced or heavily populated areas. Modern sensor networks, citizen science, and mobile measurement strategies are increasingly being employed in noise mapping; however, the paper does not provide any reflections on that.

3.2.2.3 Relation to proposed work

This study is of great importance to my research because it points out that noise is not an absolute phenomenon, but rather a complex interaction of various factors including the position of the source (e.g. road traffic) and the urban layout among others. Thus, in my mapping work, it would not only be sufficient to locate the areas of high noise, but I should also consider street orientation, height of buildings, the availability of open spaces, and different types of land uses in determining the intensity gradients across the area. The section on sampling methods in the review could guide me in the sensor location decision (grid, categorized by road classes, or straight through) as it discusses various methods that give me a good starting point. Combining the noise maps produced from my empirical work with the urban form variables, I will be able to either confirm or reject some of the authors' relationships (e.g. it is argued that noise is less where there are green spaces or building setbacks). My mapping will thus not only reflect the noise levels but also provide the local authority with planning insights that are specific to my area.

3.2.3 Noise pollution and human health

3.2.3.1 Summary

This review paper[3] discusses the detrimental impacts of noise pollution (unwanted sound in the environment) on human health. The paper states that the effects are many: loss of hearing, disturbance of sleep, heart and blood pressure stress, mental health issues, annoyance, and impairment of communication. The authors argue that noise is becoming unbearable due to expanding cities, increasing traffic,

and louder machines among others. They contend that it is a governmental duty to firstly reduce noise exposure and then protect people particularly in areas where people live, sleep or study.

3.2.3.2 Critical Analysis

This review is useful, as it gathers a plethora of recognized health issues caused by noise, making the argument about the seriousness of the problem quite strong. It is very informative and touches on a wide range of health effects. However, the review also has some drawbacks: it leans heavily on description rather than empirical evidence—it does not introduce any new experiments or data. It frequently refers to general thresholds (for example, 85 dB) but fails to provide a thorough investigation of their implications in particular local contexts (urban areas, climate types, and building types). Furthermore, it does not offer much practical advice on how to measure or map noise in a specific area, or how to set up a monitoring system.

3.2.3.3 Relation to proposed work

This review really motivates my mapping project: it proves that noise is not only an annoyance but also a factor that can harm health. Consequently, my project (mapping noise levels) can contribute significantly by indicating the worst exposure areas and thereby identifying the vulnerable population. Additionally, as the review discusses thresholds (safe versus harmful noise levels), I can incorporate those into my mapping to identify hot spots. However, since the review does not provide detailed mapping methods, my project will be an opportunity to apply sensor placement, spatial interpolation, and local calibration for the preparation of noise maps for my city and present practical results rather than just theory.

3.2.4 Mapping techniques

3.2.4.1 Summary

The work by Merchan et al. in the paper[4] investigates the performance of 15 distinct models for the purpose of the large-scale road traffic noise mapping. The researchers apply their methods on the varying grid sizes to identify how well the noise levels are predicted over the corresponding areas. Their aim is to determine the most suitable modeling technique that provides accurate noise maps at the large-scale extent. According to the authors, both the choice of the model and the grid size significantly influence the mapping quality.

3.2.4.2 Critical Analysis

This research is pragmatic because it tests numerous models and compares their performance instead of merely discussing the theory. In this way, they provide valuable insights into the models' trade-offs (complex vs simpler models, fine vs coarse grid). Nevertheless, the research might be confined to certain aspects: the accuracy claimed by the authors is influenced by the landscape, traffic conditions, and data collection methods, so the model that is "best" in their case might not apply to every urban scenario. Moreover, by only considering road traffic noise the authors exclude other sources (industrial, construction) and the study does not take into account deeply the temporal variability (such as night vs day) or the local micro-scale effects (e.g. barriers, building facades).

3.2.4.3 Relation to proposed work

This paper is of great importance to me. In my project, I will be selecting a model to either interpolate or forecast the noise levels in between the spots where noise was measured. Their comparative analysis provides me with a list of candidates (and pitfalls) to examine in my area. Furthermore, their conclusion that grid size is an important factor lets me know that I should try out various spatial resolutions to find out which one is the most suitable for my city. Since my project is based locally and most likely has a different terrain, traffic patterns, and building layouts than the one used in the study, I can take their approach as a standard but still check the models against my measured data to select the one that is best for my area. That is, I will follow their methods as a preliminary guide and then customize model selection, grid resolution, and calibration to fit my specific environment.

3.2.5 Evaluation of urban traffic noise pollution

3.2.5.1 Summary

In their research [5], Yang and co-workers generate noise maps of the Chancheng District, Foshan, China, for the whole day and assess traffic noise pollution. They categorize the district into different areas according to land usage (residential, commercial, etc.) and analyze the noise levels concerning the prescribed limits for each area. Surprisingly, it turns out that in a few areas, noise during the night is even more than that during the day, as vehicles go faster in less traffic. Moreover, they identify locations like the front of buildings where the sounds are high for a long time and also the places that are usually quiet like near the buildings where the sounds are high for a long time and also the places that are usually quiet like near the buildings and the areas that are usually cold and lonely seem to face long durations of noise that exceed the limit. They call for more noise mitigation, especially during the night.

3.2.5.2 Critical Analysis

The paper excels in using actual noise maps and matching them up with land-use categories, which ultimately reveals the areas where people are most impacted. Moreover, the note about off-peak noise levels being higher has us questioning some assumptions (it is not always the case that the busiest hours are the worst). However, the study has the limitation of being exclusively concentrated on traffic noise (while other sources such as industry, construction, etc., are excluded). Moreover, the extent to which the findings can be generalized is restricted — what works in the case of Foshan may not be automatically applicable to your city (due to differences in road types, traffic modes, building layouts). They also do not delve into micro-scale effects (e.g. shielding by walls, tree barriers) or temporal changes beyond the 24-hour mapping scenario. Finally, they recommend night when they call for mitigation but they don't go into detail about the strategies nor show cost/feasibility of interventions.

3.2.5.3 Relation to proposed work

This work holds a lot of importance. We can draw upon its method of making 24-hour noise maps from its different stages: measuring, interpolating, and then land-use zoning. The authors' discovery of more noise during non-peak hours leads me to the conclusion that I should not ignore the quiet times – mapping should be done for all times of the day. In addition, it is valuable to make comparisons of the measured (or modeled) noise with the regulatory or health thresholds (zone-wise) and I can do this for my city to point out the "hotspots". I will also be guided by their limitations: I should aim to bring in more sources (besides traffic), pay attention to more delicate spatial detail (barriers, building effects), and propose mitigation strategies that take local constraints into account. Finally, checking if off-peak noise is more than what is expected in my region could expose local patterns that are intriguing and could be used to support my work.

3.2.6 Evaluation of noise pollution in urban traffic hubs

3.2.6.1 Summary

Fiedler et al.[6] conducted their research in a major Latin American city by evaluating several busy traffic hubs to determine the level of noise in those areas using sound measurements conducted in person at different times. Their measurement was compared to environmental noise guidelines (limits) and it was found that many traffic hubs are a way above the safe noise levels. Furthermore, they have examined the mechanism of noise propagation in those hub regions indicating that the amalgamation of traffic, street configuration, and local conditions leads to high exposure. The very bad types of hubs (like intersections, main roads) are pointed out by the study and it recommends that the planners and regulators have to be

extra cautious with these hot spots.

3.2.6.2 Critical Analysis

This paper has solid ground beneath it as it relies on real field measurements in urban traffic hubs and not just computer simulations. This is one of the factors that enhance its credibility. In addition, there is no doubt that concentrating on traffic hubs is beneficial since they are the places where people are most numerous and exposed. Despite this, limitations are still apparent: the total number of measurement points could result in limited coverage and consequently interpolation in other areas may be less dependable. Furthermore, the investigation appears to be primarily concerned with traffic noise and to ignore other noise sources that have less but still significant impact (for instance, construction and human activity). The shape and location of the local environment (buildings, barriers, vegetation) is not very much incorporated or controlled through modeling in the study. Additionally, they seem not to consider temporal variability except from a few time slices, thus daily/seasonal fluctuations might be less examined.

3.2.6.3 Relation to proposed work

The article provides me with actionable experience: in my mapping project I must consider traffic hubs (intersections, main roads) as priority measurement places because they are the ones which frequently exceed the safe limits. Measured-vs-guideline comparisons will serve as a good benchmark: I can do the same for my city to find out which areas are too noisy. Moreover, I will also have to look at noise spreading locally (attenuation, barriers) around the hubs. But I will go further than their study by increasing the spatial coverage (not only hubs), considering other noise sources and by modeling the temporal variation (day, night). To sum up, their methodology will help me in planning when and where to place the sensors, how to benchmark the results and in what kind of hotspots to expect.

3.2.7 Analysing and mapping spatial and temporal dynamics of urban traffic

3.2.7.1 Summary

The study mentioned in paper [7] deals with the variability of traffic noise both in space and time, investigating a Turkish city as its case study. They performed 114 equivalent noise level (LAeq) measurements at 38 stations representing different land use types (residential, mixed residential/commercial, and industrial) during three different time periods of the day (morning, midday, and evening). They created noise maps, did area comparisons, and provided evidence that mixed residential/commercial zones had the highest average noise levels. Turkish legislation only allowed the exceeding of limits at a few places, however, when comparing with a stricter international standard, it turned out that over half of the sites were not safe.

3.2.7.2 Critical Analysis

Field measurement combined with spatial mapping during different times of the day is one of the most significant contributions of this study, as it provides a clearer view of urban noise dynamics. The classification by land use also plays a positive role in this as it helps to identify the most affected areas. However, there are some limitations that are obvious: the 38 measurement points may not be enough to capture the detailed spatial variation. Also, the temporal dimension is restricted to three time slots only, which can miss the peaks or the variation during the night. The comparison with international standards indicates that the limits set by the authorities are important, yet the paper does not go into the details of noise control measures or the modeling of noise propagation effects (for instance, barriers, terrain). Moreover, the study is limited to the city of Kahramanmaraş, so its findings may not be directly relevant in case of another city with distinct traffic, weather, or urban configurations.

3.2.7.3 Relation to proposed work

This paper presents a model that is very similar to what I am going to do: I will take measurements from different land use areas, throughout the day, and then I will map the results. Additionally, I will not only compare the local noise levels with different standards (both local and international) but also point out the areas that are safe to live in. However, I will work on their limitations: more measurement points (denser grid), night period included, and noise travel modeling (barriers, building layout) are what I am planning to do. Moreover, their study indicated that mixed residential/commercial zones had the highest levels of noise, therefore, I might consider such mixed zones in my city as possible hotspots. Theirs is an approach that guides me in sensor placement, time scheduling, and mapping methods.

3.2.8 Significance and implications of noise mapping for noise pollution control

3.2.8.1 Summary

The chapter[8] points out that noise maps are very beneficial, referring to them as “hotspot” detectors (places where noise is extremely high and, therefore, hazardous), and where measures to reduce noise and the public and policymakers can be informed. The noise map creation steps such as measuring, modeling, and visualizing noises are mentioned and how these maps can help regulate, allow urban planning and raise awareness in the community. The authors emphasize that mapping is not just about presenting figures; it is about making noise data visible and usable for the control of exposure.

3.2.8.2 Critical Analysis

This chapter is useful because it bridges theory and practice: it doesn't just explain why noise is bad, but focuses on the process and purpose of mapping. It helps the reader see how to go from measurements to actionable plans. However, it is somewhat high level: it doesn't deeply dive into the technical challenges (sensor calibration, spatial interpolation errors, local heterogeneity) or cost constraints, especially in low-resource settings. Also, while it advocates for policy uptake, it doesn't always provide detailed case studies or proofs of effectiveness in many different contexts.

3.2.8.3 Relation to proposed work

For my project, this chapter is quite helpful. It reinforces that my mapping must be not just data collection, but should aim to support decision making and regulation. I can use its flow—measure → model → visualize → communicate—as a roadmap when designing my project steps. Also, its emphasis on making noise maps understandable to policymakers and the public reminds me to include good visual design, legends, classification of risk levels, and maybe web or mobile dissemination. Because the chapter doesn't go into deep technical detail, I will need to combine it with more technical papers (for interpolation, calibration, error analysis) in my plan.

3.2.9 Noise pollution-sources, effects and control

3.2.9.1 Summary

Singh & Davar[9] examine how noise pollution arises (from traffic, loudspeakers, industry, etc.), what harms it causes, and how it might be controlled. They did a cross-section survey in Delhi to identify which noise sources bother people most, and they report that automobiles and loudspeakers are major culprits. They list effects such as interference with communication, insomnia, lowered efficiency, and in severe cases hearing loss or mental stress. They also discuss control measures: better machine design, soundproofing, enforcing noise laws, and public awareness/education.

3.2.9.2 Critical Analysis

The article provides a very basic but at the same time comprehensive reference about the topic: what noise pollution is, who are the people suffering from it, and the general means of its reduction. It is particularly useful for the novice reader. Nevertheless, there are some drawbacks to it. The empirical part (the survey) is very shallow and limited (the research concerns only Delhi). It does not offer any strong quantitative modeling or spatial mapping. The proposals for the control measures are quite general and idealistic — there is virtually no information on the costs, feasibility, or how to prioritize

actions. Furthermore, due to its age (2004), it has not taken into consideration the application of modern technologies like sensor networks or GIS mapping techniques.

3.2.9.3 Relation to proposed work

This paper is a good source of information and motivation for my project: it is useful to know what the common sources are (traffic, loudspeakers) for deciding what to monitor. Their mentioned health effects tip the scales towards mapping (to illustrate the danger zones). The control measures indicated by them (like sound insulation or stricter enforcement) open up the way for what to suggest later after mapping. However, the paper does not provide the mapping methodology, so I will have to merge its ideas with the more recent technical literature (on spatial interpolation, sensor deployment, noise propagation). At the same time, I will be able to check if the sources they mention (traffic, loudspeakers) are truly the most significant ones in my city, or if there are local peculiar sources that should be included.

3.2.10 Noise pollution and cardiovascular problems

3.2.10.1 Summary

This[10] is a review article that looks at how transportation-related noise (from roads, railways, aircraft) affects heart and blood vessel health (cardiovascular disease). The authors collect evidence from many epidemiological studies showing that people exposed to more transportation noise have higher risks of hypertension, heart disease, and mortality. They also go deeper into mechanisms: noise at night disrupts sleep, increases stress hormones, leads to oxidative stress and inflammation in blood vessels, and can disturb metabolic pathways. They propose that noise should be considered a cardiovascular risk factor, not just an annoyance.

3.2.10.2 Critical Analysis

This review is extremely current and very thorough—not only does it imply just that “noise is bad,” it actually goes on to demonstrate the ways that noise can cause diseases by tracing biological pathways (stress, inflammation, oxidative damage) and so forth. It has made a very strong argument for taking noise seriously from the point of view of public health. On the other hand, there are a few drawbacks: a lot of the epidemiological studies are observational, which means that there can be confounding factors (e.g., other environmental exposures, socioeconomic status) that are difficult to control for completely. There are times in the review when it appears to rely mostly on animal or tiny human experimental studies to establish mechanisms, which may not be perfectly appropriate for human scenarios at all. Furthermore, while giving the example of mitigation strategies (traffic management, barriers, insulation), the review falls short in terms of discussing how their implementation would be in terms of being

affordable or potentially effective in many cities, especially in those that are less developed.

3.2.10.3 Relation to proposed work

This article is very relevant for me because it strengthens the health justification of my mapping project: if noise can contribute to heart disease, then mapping where noise is worst becomes more important for public health. In my project, I can use the thresholds or exposure levels reported in this paper (or cited studies) to define "high risk" zones in my noise maps. Furthermore, when I present my maps to stakeholders or city planners, I can refer to these mechanistic and epidemiological links to argue for interventions in the hotspots my map reveals. Also, because the review highlights night noise as especially harmful (for sleeping and physiological stress), I should make sure my mapping includes nighttime measurements, not only daytime.

3.2.11 Assessment of Noise Pollution Indices in the City of Kolhapur, India

3.2.11.1 Summary

Hunashal & Patil[11] measured daytime noise levels across five "critical zones" in Kolhapur, India (educational, commercial + residential, industrial + residential, etc.) to assess how bad noise is in different parts. They computed various noise pollution indices (like Leq, noise pollution level, exceedance, etc.) and compared the values across zones. The study found that noise levels in many zones were higher than the acceptable limits, indicating serious noise pollution in Kolhapur.

3.2.11.2 Critical Analysis

The paper provides valuable information through its methodology of measuring the noise pollution at different places and applying many indices instead of just one and thus getting a more detailed view of the situation. Moreover, examining the issue of noise in practical urban zones during the day makes it relevant. However, there is a drawback in the sense that it only analyzes daytime which means the night noise behavior (which could be vital) is not considered. The area covered is also limited to the "critical zones" which indicates that a large part of the city is still unassessed in terms of noise pollution. Moreover, the authors predominantly present descriptive comparisons, and do not truly model the propagation of noise or the impact of urban form (buildings, barriers) on it. Besides, the authors' suggestions are too general (for example, it is better to control noise in the areas with problems) rather than providing specific mitigation plans.

3.2.11.3 Relation to proposed work

This piece of research is beneficial to me in various aspects: first, I can utilize their method of splitting the city into zones (residential, commercial, mixed) and calculating several indices besides one. Their observation that most areas are laden with pollution has already led me to consider the same polluted places in my city. Besides, I have to carry out my measurements both at night and during the day (not only daytime) to gain a comprehensive understanding, for instance, their limitation is lack of data at night. While designing my sensor placement, I can adopt their critical zones concept to guarantee that I have collected data from all the important land-use types. Finally, I can apply their thresholds and indices as standards for my interpretation of the results, but at the same time, I can make them even better by incorporating spatial interpolation, propagation modeling, and possibly night coverage.

3.2.12 Modelling Of Traffic Noise Pollution

3.2.12.1 Summary

The research paper [12] examines the modeling of traffic noise in Coimbatore, a city in India. The researchers took sound levels along a part of a national highway (NH-209) at various traffic and weather conditions. Amongst their recordings were hourly traffic volume, average speed of vehicles, temperature, humidity, and surface temperature. Subsequently, the authors combined the data from traffic and environmental conditions using a regression analysis to develop a mathematical prediction model that converts the noise level (in dB) under those conditions. The R^2 values for the regression varied from about 0.1 to 0.7 according to the data used.

3.2.12.2 Critical Analysis

This article is of a high value because it makes an effort to connect the observable environmental and traffic factors with noise levels in a real case and that is just the data needed in a predictive mapping project. The addition of weather factors (temperature and humidity) is a positive aspect since sound transmission is weather-dependent. Besides, the application of both Leq and L10 allows for a more subtle interpretation (not merely the average). Nevertheless, there are some drawbacks. The R^2 value is rather small in some instances (0.1 is quite weak), indicating that the model does not account for all the important variations. The research has been done only in one specific highway area, therefore it can't be easily applied to different types of cities (busy streets, tiny lanes, building reflections, etc.). It also doesn't deal with the issue of spatial interpolation (moving from point predictions to area maps) or micro-scale features such as barriers or building shielding. Moreover, the temporal coverage might also be restricted (they took data at several times but may not have covered the entire day/night cycle in

detail).

3.2.12.3 Relation to proposed work

This work is quite relevant for me. I can adopt a similar regression modeling approach: gather traffic and environmental data along with noise measurements, then build a model to predict noise in unmeasured locations. Their approach shows which variables might be useful (vehicle count, speed, temperature, humidity) — I should include those in my model too. But because their model has limitations, I'll need to extend it: use more predictors (such as road geometry, building distances, barriers), choose points across different urban forms (not just highways), and test different model types (e.g. more advanced regression, machine learning). Also, after I have a predictive model, I will need to spatially interpolate or map it across the city (e.g. using kriging or GIS) to produce continuous noise maps. Their work gives a base, but my project can expand and refine it for local conditions.

3.2.13 Directions in Strategic Environmental Noise Mapping

3.2.13.1 Summary

This paper[13] reviews how "strategic noise mapping" (SNM) has evolved in Europe since the the Environmental Noise Directive (END) was introduced in 2002. The authors discuss improvements in methods, especially the adoption of CNOSSOS-EU, which is a standardized noise calculation method used across European countries for consistency.They also examine how technological advances—such as better GIS tools, remote sensing, bigger computing power, and possibly participatory / citizen sensing—are shaping future directions. The paper argues that strategic noise maps are not just academic but are essential tools for policymakers, helping to prioritize noise reduction, inform exposure assessments, and guide actions to protect public health.

3.2.13.2 Critical analysis

The article in question effectively tells the story about noise mapping in a very broad way, showing its present and future all the way to the top of the policy-making and strategic level. The mention of CNOSSOS-EU is particularly relevant because the application of a single method across the board minimizes discrepancies arising from different nations. The introduction of future technologies such as GIS, sensors, and crowdsourcing brings the discussion to the future. Nevertheless, the article has its shortcomings: it is primarily Eurocentric (adhering to EU conditions, regulations, and infrastructure), consequently, most of the concepts might not be easily adopted in countries with completely different settings. Furthermore, as a review, it generally stays at a higher level and does not always interact deeply with the hardships in the periphery (limited budgets, scarce data, difficulties in calibrating, unusual local

geography) of the less developed regions. The authors present plenty of promising paths in their paper, but very often they do not offer detailed ”how to” steps for execution in low-resource settings.

3.2.13.3 Relation to proposed work

This review is of a very high relevance for me, since my project is literally a ”strategic noise mapping” exercise on a local scale. I gather from this paper the necessity to employ or adjust a standardized noise model (in the case of CNOSSOS-EU) so as to attain results that are reliable and can be compared. Besides this, the paper also proposes the adoption of modern tools (GIS, computing at a better level, and possibly participatory input) which can be used in my project to enhance data quality and public involvement. The point that maps should be the basis for policy means that I have to create my maps not only as pieces of academic work but also as the tools city planners, decision-makers, or public health officers can use. Since a lot of their ideas are drawn from high-resource settings, I will have to make adjustments based on my local limitations (for instance, fewer sensors, limited budget). To sum it up, the paper provides me with a tactical planner: apply good modeling standards, utilize technology, and concentrate on the usability by stakeholders.

3.2.14 Data collection using wireless sensors

3.2.14.1 Summary

This chapter [14] discusses using wireless sensor networks (WSNs) to monitor noise pollution in cities. The authors compare different data collection protocols (specifically, CTP — Collection Tree Protocol — and DMAC) in terms of energy consumption, reliability, and suitability for continuous noise monitoring. They conduct experiments and show that CTP with the low power listening (LPL) option offers a good balance between power use and data delivery reliability for noise sensor networks.

3.2.14.2 Critical analysis

I personally consider this chapter very practical and useful: it tackles a specific problem of energy consumption when deploying multiple sensors in the field, instead of only concentrating on measurement theory. The protocol comparison indicates that the writers are cognizant of the actual limitations, such as battery life and network traffic. On the other hand, there are some drawbacks; the study is limited since only two protocols are tested, which might not include all the issues that can occur in real-world applications (sensor failures, radio interference, maintenance, calibration drift). Moreover, the authors don’t delve into how the network can be scaled up in extremely high-density metropolitan areas with so many nodes, nor do they tackle the issues of ensuring spatial coverage or of fault tolerance. In addition,

the chapter does not cover the pathways to integrating sensor data into spatial noise models or to GIS technology for the production of full noise maps.

3.2.14.3 Relation to proposed work

This chapter is quite relevant to my project because I will need to deploy many noise sensors in a city, and one big constraint is power / energy and data communication. The lesson that CTP + LPL gives a good tradeoff helps me choose which communication protocol or sensor network architecture to adopt. In my mapping, I'll need to think not just about where to place sensors but also how to ensure they keep working over time, how to collect data reliably, and how to manage power. But I will also need to complement this with mapping strategies: after collecting data, I'll need to do interpolation, spatial modeling, and visualization, which this chapter doesn't cover. In short, this work helps me with the "sensor network / data collection" side of my project.

3.2.15 Crowwdsensing approach to Noise pollution monitoring

3.2.15.1 Summary

This paper[15] introduces a crowdsensing platform designed to monitor noise pollution in real time within cities. They use data collected from many distributed participants (citizens or mobile devices) to continuously sense noise, aggregate it, and analyze spatial and temporal patterns. The platform allows mapping of noise levels, spotting hotspots, and supporting decision making in a smart-city context.

3.2.15.2 Critical analysis

This work is interesting because it leverages crowdsourcing—using many ordinary devices rather than fixed expensive sensors—to scale noise monitoring. That helps with coverage and cost. Also, doing real-time data collection and analysis is very useful in dynamic urban contexts. But there are challenges: the data quality from many consumer devices might vary (calibration, device noise, orientation), and spatial bias may occur (some areas have more participants than others). Another limitation is privacy, participation incentives, and ensuring continuous coverage. The paper may not deeply solve how to deal with biased sampling or missing data in less populated or less connected zones.

3.2.15.3 Relation to proposed work

This article is a good source of information for my project. A crowdsensing method could enable me to monitor more locations without the need for a large number of fixed sensors to be installed, especially in the case of limited-budget areas. The data from the crowdsensing will be treated together with the data

from the fixed sensors for validation and calibration. Further, monitoring in real-time would mean that I could detect temporal changes (rush hour peaks, special events) which would add more value to my maps. However, I would have to deal with issues like calibrating device data, managing inconsistency in participation, and making sure there is a steady coverage in all neighborhoods (not just the ones with a lot of people). It would be advisable to think through participant recruiting, the handling of noisy or biased data, and integrating of crowdsourced measurements with spatial interpolation/modeling.

3.3 Literature Review Summary Table

Table 3.1: Research Summaries

Author	**Method**	**Results**	**Limitations**
Gupta et al. [1]	Literature review and synthesis of epidemiological studies	Noise affects hearing, sleep, cognitive physiological health in children and other age groups	Doesn't present new empirical data or local modeling
Morillas, Gozalo, González, Moraga et al. [2]	Conceptual review linking urban planning and noise propagation	Urban form, land use, street layout strongly influence noise exposure	Focused on theory and not detailed on implementation in low-resource settings
Jariwala et al. [3]	Review of health effects of noise pollution	Noise causes hearing loss, sleep disturbance, cardiovascular stress, annoyance	Lacks spatial / mapping methodology and local case studies
Merchán et al. (2013) [4]	Comparative test of 15 mapping models at various grid resolutions	Model and grid choice strongly affect mapping accuracy	Results may not generalize to all urban contexts

Yang et al. [5]	Created 24-hour noise maps by zone and compared with limits	Some off-peak noise levels exceed rush-hour levels in zones	Only traffic considered; limited micro-scale detail
Fiedler et al.[6]	Field measurements at traffic hubs and mapping	Many hubs exceed safe noise thresholds	Limited spatial coverage and doesn't include other noise sources
Doygun And Kuşat Gurun (2008)[7]	Measurements at 38 sites across times + mapping	Mixed residential/commercial zones had highest noise; many points exceed limits by international standards	Only three time periods, moderate number of sensors
Kumar, Chauhan & Garg[8]	Review on role of noise mapping Maps	help identify hotspots and support noise control policy	High level, does not deeply tackle technical challenges
Singh & Davar (2004)[9]	Survey in Delhi	Found vehicles and loudspeakers as major noise sources affecting health.	Limited area, lacks data modeling and modern techniques.
Münzel et al. (2021)[10]	Review of mechanistic studies	Noise increases cardiovascular disease risk via sleep disruption, stress etc	Mostly observational evidence and limited causal proof
Hunashal & Patil (2012)[11]	Measured noise levels and calculated pollution indices in city zones.	Most zones exceeded noise limits, showing high pollution.	Only daytime data and limited spatial coverage.
Coimbatore Study (IJERA, 2012)[12]	Regression modeling using traffic and weather data.	Built equation linking vehicle flow, weather, and noise levels.	Weak model fit (low R^2) and only highway-based data.
Murphy, Faulkner & Douglas (2020)[13]	Review of European strategic noise mapping methods.	Highlighted CNOSSOS-EU and new tech trends for mapping.	Focused only on EU context; lacks practical local examples.
Filipponi, Santini & Vitaletti (2008)[14]	Compared wireless sensor protocols .	Found CTP with LPL gives best balance of power and reliability.	Small-scale test; doesn't cover large urban networks or GIS use.
Jezdović et al. (2021)[15]	Built a crowdsensing platform	. Created real-time maps showing city noise hotspots.	Data quality and coverage depend on user participation and device accuracy.

3.4 Conclusion

After reviewing all fifteen research papers, it is clear that noise pollution is a serious and growing problem that affects both the environment and human health. Most studies agree that long-term exposure to

noise can cause hearing loss, stress, sleep problems, and even heart-related diseases. Many researchers have used methods like field surveys, GIS-based mapping, and machine learning to measure, predict, and visualize noise levels in cities. These studies show that traffic and industrial areas are the main sources of harmful noise. However, most of the existing work focuses on specific cities or uses limited datasets, making it hard to generalize results. Therefore, there is still a need for more localized and continuous noise monitoring systems. My proposed noise pollution mapping project aims to fill this gap by developing a local, data-driven approach to visualize and analyze noise levels, helping communities understand and manage noise pollution more effectively.

Chapter 4 Software Requirement Specifications

This section delivers a comprehensive overview of the project's functional and non-functional requirements, use cases, GUI Interface and risk analysis.

4.1 List of Features

List all important features of your system here.

- . IoT-Based Real-Time Noise Monitoring
- . Geospatial Data Integration
- . Dynamic Noise Pollution Mapping
- . Data Storage and Management
- . Analytical and Statistical Tools
- . User-Friendly Web Interface
- . Decision Support System
- . Pilot Deployment and Evaluation

4.2 Functional Requirements

The functional requirements are given below.

- The system allows user log in (e.g., admin, city planner, public) with correct credentials to see authorized features.
- IoT-based sensors constantly stream real-time noise level values (in decibels) from various city locations.
- Sensor-collected data is securely sent to the central server/database through IoT communication protocols.
- All sensor data received are stored in a database of well-structured formats for real-time and history analysis.
- The system shows current noise levels on an interactive GIS-based map with color-coded zones marking intensity.
- Trends and patterns of noise pollution are analyzable by users using graphs, charts, and statistical

functions. Reports can be generated for specified time intervals or areas.

- The system notifies the user of the occurrence of noise levels exceeding a threshold in specific zones.
- Data can be filtered by location, date, or noise intensity range in order to get targeted insights.
- The system provides access to the history of noise data so that the current and historical levels of noise can be compared.
- Admin can manage user roles, sensor configurations, data sources, and system performance overall.
- A basic dashboard version for generic users to view live noise maps and general awareness data.

4.3 Quality Attributes

- **Reliability**: The system must operate continuously and properly collect and display noise information without frequent failure or shutdown.
- **Usability**: The web interface and dashboard must be user-friendly, intuitive, and easy to understand for different classes of users (public, planners, and administrators).
- **Performance Efficiency**: The system should have the capability to efficiently process large volumes of sensor data, offering real-time visualization and fast response to user actions.
- **Scalability**: EchoMap should be horizontally scalable through the addition of sensors or locations without affecting performance or major redesign.
- **Interoperability**: The system should also be capable of interfacing with other external APIs, for example, the weather information or other smart city infrastructures, in future implementations.
- **Reusability**: Similar environmental monitoring projects should be benefited by the reusability of modules like data visualization or analytics.
- **Flexibility**: The system should possess such flexibility that it can accommodate the new requirements, i.e., adding sensors, new data processing modules, or visual layers of the map, etc.
- **Responsiveness** : The user interface should be capable of providing maps, analytics, and reports within seconds to facilitate the user's interaction easily.
- **Testability** : Any module must be tested independently to ensure its functionality, performance, and compatibility with the other modules.

4.4 Non-Functional Requirements

Non-Functional requirements are listed below.

- **Performance**: The system is required to process as well as display the real-time sensor data on the map, within a period of 5 seconds after the data has been received.
- Scalability: The system should be capable of adding hundreds of sensors and keeping on processing large amounts of data without resorting to performance loss.
- **Availability**: The system is expected to be operational for at least 99% of the time, in order to maintain uninterrupted data collection and visualization.
- **Security**: It should be ensured that the communication between sensors, servers, and users is encrypted (e.g., HTTPS, MQTT over TLS), so that unauthorized access is prevented.
- **Maintainability**: The architecture of the system as well as the codebase should be segmented into modules in a manner that makes updating, debugging, and adding functionality very easy.
- **Portability**: The system needs to perform well on different browsers and operating systems (Windows, macOS, and Linux). The web interface should also be mobile-friendly.
- Accuracy: The IoT sensors need to have the capability of measuring noise levels with an accuracy of ±1 dB in order to provide good and reliable analytical results.
- **Data Integrity**: The system is required to provide a guarantee that no alteration will happen to the data during the processes of transit, storage, or retrieval.
- **Compliance**: The system needs to conform to the standards of relevant environmental data and ethical data treatment practices as per WHO or local environmental laws.

4.5 Assumptions

- **Stable Internet Connectivity**: The master server and all IoT sensors will have stable internet connectivity for real-time data exchange.
- **Proper Sensor Calibration**: Noise sensors which are used in the system are properly calibrated and provide accurate sound level measurements.
- **Limited Geographical Area (Pilot Phase)**: Installation and testing will initially be done in a specific area or city, but not on a national scale.
- **Power Supply Availability**: The IoT sensors will be supplied with a continuous power supply

(battery or power) to run around the clock.

- **GIS Data Availability**: Proper geography and map data (city maps, coordinates, and zones) will be available for GIS integration.
- **Availability of Open-Source Tools**: Open-source IoT and GIS tools and frameworks (such as Leaflet, Node-RED, or QGIS) will be used by the project, which are freely accessible.
- **Data Privacy Compliance**: The information collected is all environmental (not personal), and thus ethical and privacy concerns are non-existent.
- **Environmental Stability**: Outside factors like weather or temporary building construction noise will have little effect on general data reliability in longer-term analysis.

4.6 Use Cases

We have identified the following use cases for our web application with the assumption that the application does not malfunction.

Table 4.1: Record Noise Data

Name	Record Noise Data		
Actors	User, System (Microphone, GPS)		
Summary	The user records environmental noise using the device microphone. The system captures noise level, location, and timestamp.		
Pre-Conditions	The user must be logged in and must have microphone and location access enabled.		
Post-Conditions	Noise data, location, and time are stored in the database for analysis.		
Special Requirements	Internet connection and active GPS signal.		
Basic Flow			
Actor Action		**System Response**	
1	User opens record noise option.	2	System activates microphone and GPS.
3	User starts recording noise.	4	System collects decibel reading and coordinates.
5	User stops and saves the recording.	6	System uploads noise data to the database.
Alternative Flow			
A1	Microphone permission denied.	A1-1	System prompts user to allow microphone access.
A2	No GPS signal available.	A2-1	System displays message to enable location services.

Table 4.2: Classify Noise Type

Name	Classify Noise Type		
Actors	System (Machine Learning Model)		
Summary	The system uses a trained machine learning model to classify the recorded sound into categories like traffic, human, or construction noise.		
Pre-Conditions	Noise data must be recorded and stored in the database.		
Post-Conditions	Noise type label is added to each stored noise entry.		
Special Requirements	Trained machine learning model accessible to the system.		
Basic Flow			
Actor Action		**System Response**	
1	User uploads or records sound.	2	System extracts features using signal processing.
3	-	4	System classifies noise using trained model.
5	-	6	System stores classified noise label in database.
Alternative Flow			
A1	Model file not found.	A1-1	System displays error: “Model unavailable.”
A2	Low audio quality.	A2-1	System marks classification as uncertain.

Table 4.3: View Noise Map

Name	View Noise Map		
Actors	User, Researcher		
Summary	User views an interactive map displaying noise levels at different locations using color-coded heatmap visualization.		
Pre-Conditions	Noise data with location coordinates must be available in the database.		
Post-Conditions	User can visualize noise levels and navigate the map.		
Special Requirements	Stable internet connection and browser with JavaScript enabled.		
Basic Flow			
1	User selects “View Map” option.	2	System loads noise data from the database.
3	User zooms or drags the map.	4	System updates area and displays nearby data.
5	User clicks on a location.	6	System shows detailed noise data for that spot.
Alternative Flow			
A1	No data available for area.	A1-1	System displays “No readings found” message.
A2	Slow internet connection.	A2-1	System loads map in low-quality mode.

Table 4.4: View Time-Lapse Visualization

Name	View Time-Lapse Visualization		
Actors	User, Researcher		
Summary	Displays a time-lapse animation showing how noise levels change over time in a specific area.		
Pre-Conditions	Historical noise data must exist for selected area and time period.		
Post-Conditions	User views a visual playback of noise variation over time.		
Special Requirements	JavaScript-enabled browser, fast internet connection.		
Basic Flow			
1	User selects time-lapse option on map.	2	System fetches historical noise data.
3	User selects time range.	4	System shows changes in noise intensity.
Alternative Flow			
A1	Insufficient data for selected range.	A1-1	System shows “Not enough data” message.

Table 4.5: Generate Report

Name	Generate Report		
Actors	Researcher, Administrator		
Summary	The system generates a report showing average noise levels, most common noise types, and location-based summaries.		
Pre-Conditions	Sufficient data must exist in the database for report generation.		
Post-Conditions	A downloadable report is produced in PDF or CSV format.		
Special Requirements	Export and chart libraries installed.		
Basic Flow			
1	User selects "Generate Report".	2	System retrieves required data from database.
3	User selects report type.	4	System generates charts and summaries.
5	User downloads report.	6	System saves and exports report as file.

Table 4.6: Manage Users and Data

Name	Manage Users and Data		
Actors	Administrator		
Summary	Administrator manages users, removes invalid entries, and ensures data accuracy in the system.		
Pre-Conditions	Administrator must be logged in with proper access rights.		
Post-Conditions	Database and user records are updated successfully.		
Special Requirements	None		
Basic Flow			
1	Admin opens management dashboard.	2	System displays user and data tables.
3	Admin edits or removes records.	4	System updates changes in database.
Alternative Flow			
A1	Admin has insufficient privileges.	A1-1	System restricts access and logs event.

4.7 Hardware and Software Requirements

The hardware and software requirements are listed below.

4.7.1 Hardware Requirements

- Microphone, GPS, Internet
- . 8GB RAM, i5 CPU, SSD 256GB

4.7.2 Software Requirements

- Python, JavaScript
- Librosa, TensorFlow, scikit-learn
- Leaflet.js, Mapbox
- MySQL/Firebase

- Flask/Django
- Heatmap, Time-lapse Dashboard

4.8 Graphical User Interface

GUI dumps of each screen, with reference to the users are given below.

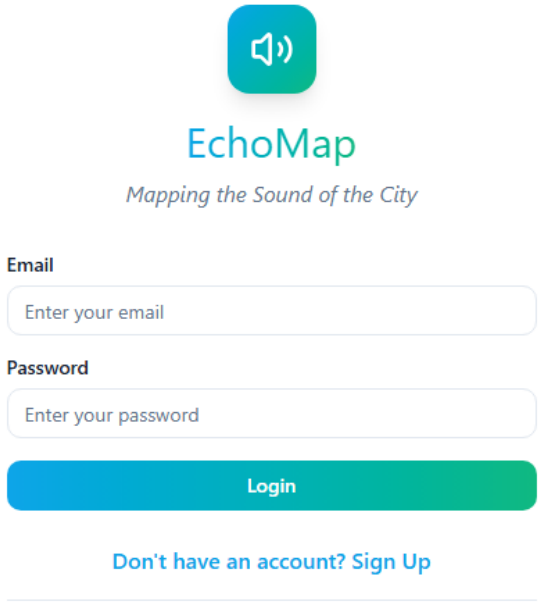

Figure 4.1: Login Page

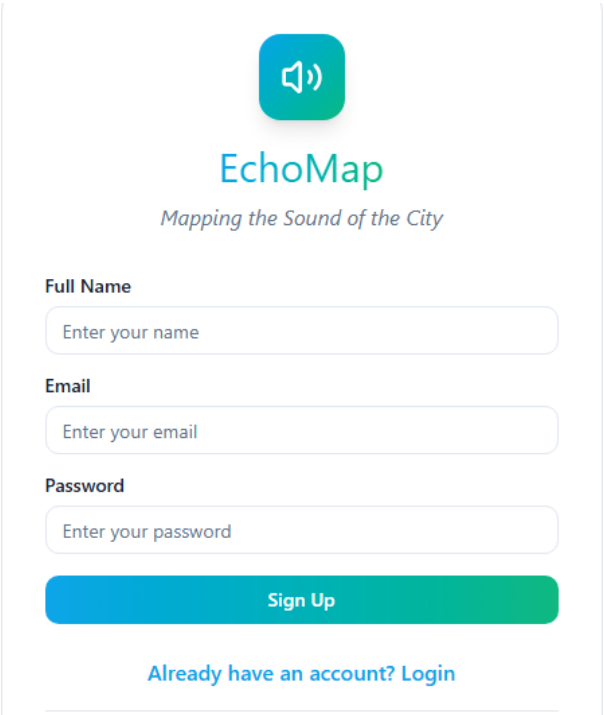

Figure 4.2: Sign Up page

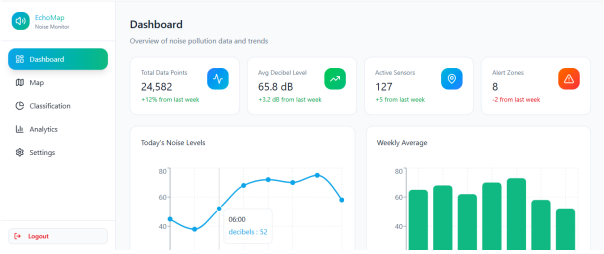

Figure 4.3: Dashboard

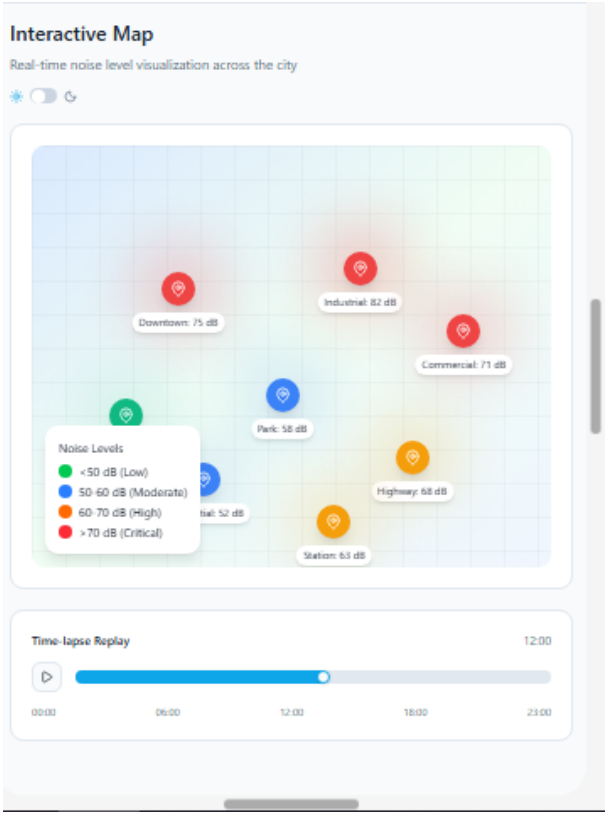

Figure 4.4: Map

Figure 4.5: Classification

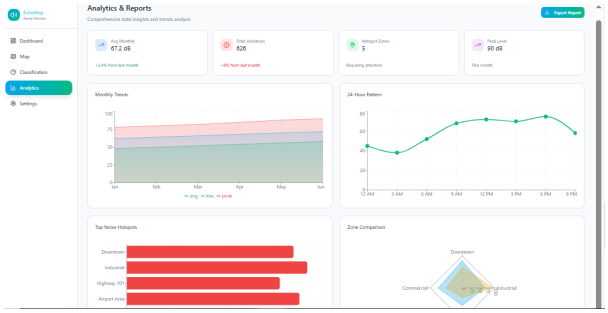

Figure 4.6: Analytics

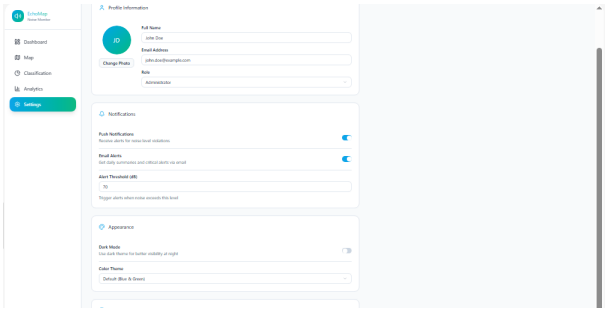

Figure 4.7: Settings

4.9 Database Design

4.9.1 Data Dictionary

The following is the data dictionary of the project.

Table 4.7: Table 4.11: Data Dictionary

Entity	Attribute(s)	Data Type	Nullable	Description
User	userID, userName, email, password, role, contact	Integer, String, String, String, String, String	No, No, No, No, Yes, Yes	Stores registered user information including name, email, and role (User or Admin).
Login	email, password	String, String	No, No	Credentials used by users for authentication and login to the system.
NoiseData	noiseID, userID, decibelLevel, location, timestamp, audioFilePath	Integer, Integer, Float, String, DateTime, String	No, No, No, No, No, Yes	Contains noise measurements recorded by users with time and location details.
NoiseClassification	classID, noiseID, noiseType, confidenceScore	Integer, Integer, String, Float	No, No, No, Yes	Stores classification results for recorded noise (e.g., traffic, construction, human).
MapVisualization	mapID, areaName, averageNoiseLevel, coordinates, timeRange	Integer, String, Float, String, String	No, No, Yes, Yes, Yes	Represents the data displayed on the noise map and time-lapse visualization.

4.10 Risk Analysis

4.10.1 Business/Operational Risks

- Resource Constraints: Lack of time, budget, or personnel may be able to slow down system development or testing.
- Dependency on External APIs: Any restriction or breakdown of third-party APIs (Mapbox, Google Maps) can hinder functionality.
- Stakeholder Engagement: Challenges in accessing the government or institution's cooperation for original data.

4.10.2 Technical Risks

- Inaccurate Measurements: Smartphone microphones can record unreliable decibel readings.
- Model Misclassification: The ML model might misidentify categories of noise using inadequate

Table 4.8: Table 4.11: Data Dictionary

Entity	Attribute(s)	Data Type	Nullable	Description
Report	reportID, generatedBy, dateGenerated, reportType, filePath	Integer, Integer, Date, String, String	No, No, No, No, Yes	Stores generated reports on noise trends and levels for different locations.
Admin	adminID, name, email, password	Integer, String, String, String	No, No, No, No	Stores administrator credentials and details for managing system data and users.
Alerts	alertID, userID, message, timestamp	Integer, Integer, String, DateTime	No, No, No, No	Stores system-generated notifications and alerts for users.
LocationData	locationID, latitude, longitude, areaName	Integer, Float, Float, String	No, No, No, Yes	Stores geographical coordinates and area names linked to noise recordings.
Feedback	feedbackID, userID, message, date	Integer, Integer, String, Date	No, No, No, No	Stores user feedback and suggestions about system performance or issues.

or noisy training data.

- Integration Failures: Issues may occur in mobile, backend, and visualization piece integration.
- Performance Bottlenecks: System performance might slow down with big data or multiple users.
- Tool Compatibility: Library, API, or framework incompatibilities might hinder development.

4.10.3 Environmental Risks

- Testing Limitations: Weather, ambient noise, and other environmental factors may taint real-world data accuracy.
- Hardware Failure: Field data collection is hindered by device or sensor malfunction.

4.11 Conclusion

The SRS or Software Requirement Specifications for EchoMap specifies all requirements and measures for the development of a system that can probably be called modular, three-tiered, and such a system able to afford the recording, classifying, and visualizing of environmental noise data by the user through web and mobile platforms. Recording noise data, classifying sound types with the help of machine

learning, viewing heatmaps and time-lapse visuals interactively, creating analytics reports, and controlling users and data are some of the system's detailed use cases. The GIS tools Leaflet.js and Mapbox are incorporated into the system, while Python, TensorFlow, and Librosa are used for backend processing. Mobile phones with GPS, regular computing systems are the minimum hardware requirements, and the database design for user information, noise data, classifications, and reports is built to support them. The risk analysis conducted covers operational, technical, and environmental problems, among others, limited resources, inaccurate measurements, and integration or performance problems. To sum up, the SRS guarantees that EchoMap is going to be a scalable, data-driven platform for urban noise pollution monitoring and analysis that is provided with the necessary efficiency.

Chapter 5 Proposed Approach and Methodology

The supposed method and process will explain the proposed plan and systematic process in- involved in developing the application.The suggested method and process will elaborate on the recommended plan and systematic approach in-were involved in developing the application.

5.1 Introduction

The intended approach and methodology for EchoMap, a smart noise pollution mapping and analysis system, are addressed in this chapter. The goal is to develop a system combining data harvesting, preprocessing, machine learning–driven classification, and interactive visualization. The algorithmic and procedural aspects of the system are discussed in the following sections, outlining how various modules interact to decompose raw environmental noise into meaningful insights.

5.2 Overview

EchoMap combines data science, cartography, and machine learning into one workflow. The entire process takes audio from mobile phones of the users, processes it on a powerful backend server, identifies noise sources through pre-trained artificial intelligence models, and shows the output as a heatmap with user interaction on either a web or mobile platform. This strategy promotes scalability, availability, and productivity, at the same time limiting hardware costs to a minimum.

5.3 Data Collection Framework

The first step involves the capture of ambient sound samples with smartphone microphones. Each sound sample is labeled automatically with metadata such as timestamp, GPS coordinates, and decibel level (dB). Data is transmitted securely to the central server to process. To ensure consistency, all devices follow a calibration protocol reducing variations based on differences in microphone sensitivity.

5.4 Data Preprocessing and Feature Extraction

Raw sound recording is processed using audio signal processing techniques. Noise samples are first normalized, denoised, and segmented into fixed-length time frames. Then, audio features such as MFCC (Mel-Frequency Cepstral Coefficients), spectral centroid, zero-crossing rate, and chroma features are extracted using the Librosa library. These features are then used as inputs for machine learning classifiers.

5.5 Noise Classification Using Machine Learning

Here, the learned features are transferred to supervised machine learning models in order to learn the noise type (e.g., human speech, light traffic, heavy traffic, construction, etc.). Various algorithms such as Support Vector Machines (SVM), Random Forest, and Convolutional Neural Networks (CNN) are tried out. After testing, the model with the best accuracy and lowest overfitting rate is chosen for deployment.

5.6 Model Training and Validation

Training is performed over a labeled set of city sounds of different diversity. The set may include open repository samples like UrbanSound8K and locally recorded samples. Training (70%), validation (15%), and test (15%) sets divide the data. Standardization and augmentation strategies address imbalance and improve robustness across environments.

5.7 Backend Processing Pipeline

After training, the classification model becomes part of the backend pipeline which is developed using Python (Flask or Django). The pipeline performs the following roles: receiving requests from the mobile app, applying the prediction algorithm to the input sound samples, saving the outcomes in a database, and readying the data for visualization. The use of asynchronous task management gives the system a performance that is both smooth and scalable regardless of the amount of traffic.

5.8 Data Storage and Management

All the data that has been collected and processed, which includes sound samples, classes, timestamps, and locations, is organized in a structured database (for example, PostgreSQL). A clear schema is used to support effective searching and getting back data. Historical data is kept in storage at regular intervals, and privacy measures are put in place to disguise users' data before it is processed.

5.9 Visualization and Mapping Module

EchoMap generates a visual display in the form of an interactive heatmap that is created with the help of Leaflet.js or Mapbox. The noise levels in different areas are represented through the varying intensity of colors, and at the same time, the main noise types are depicted using icons or markers. The users also have the option to filter the data based on the time (day/night mode) or the category, which would help them to grasp the distribution patterns of noise even more clearly at the specific locations.

5.10 Temporal Analysis and Time-Lapse Feature

EchoMap's time-lapse feature is one of its most striking characteristics. Users have the option to see how the noise levels have changed for a whole day or even a week. The system not only produces real-time noise patterns but also helps in finding the noisy periods through linking timestamps with geospatial data.

5.11 System Integration and User Interface

The simplicity and ease of interaction characterize both the web and mobile frontend interfaces. Displaying real maps, searching for areas, and noise trends interactivity are among the features of the UI. APIs create the links between the frontend and the backend that allow for real-time data access, aiming at communication without obstacles and data synchronization.

5.12 Evaluation Metrics and Testing

The evaluation of the system is performed using accuracy, precision, recall, and F1-score metrics among the classification performances. The usability, responsiveness, and the accuracy of the maps provided by the system are also evaluated through pilot studies in controlled cities like university campuses. The trustworthiness of the results is further improved by including real-time user feedback.

5.13 Security, Ethics, and Data Privacy

Heavy-duty confidentiality measures have been implemented as the system handles user-uploaded sounds and geographical information. Only noise levels (not private conversations) are preserved in the database. Encryption combined with anonymization makes it impossible to identify any person through the data left behind. Ethical rules are in compliance with the principles of environmental data collection.

5.14 Conclusion

The proposed methodology offers a robust, scalable, and data-demanding framework for smart noise monitoring. Through combining of mobile data gathering, classification powered by machine learning, and interactive mapping, EchoMap closes the loop between technology and environment awareness. The methodology not only addresses actual environmental issues of noise pollution but also sets the stage for future smart-city integrations and environmental decision-support tools.

Chapter 6 High level and low level design

The details of high-level and low-level design of the system we are provided in this chapter.

6.1 System Overview

Echomap software comprises the following key modules:

6.1.1 General Description

EchoMap is an intelligent, web-enabled noise pollution monitoring and visualisation platform that records, classifies, and displays environmental noise data in real time. It takes advantage of the sensing capability of smartphones or mobile phones to measure decibel readings, geographic location (GPS coordinates), and timestamps. These data are processed with a cloud-based backend that performs machine-learning-based noise classification and produces visual analytics like interactive heatmaps and temporal trends.

6.1.2 Functional Overview

The primary operation of EchoMap falls into the below-mentioned major modules:

6.1.2.1 Data Collection Module

- Captures environmental sound using the microphone of the smartphone.
- Captures related metadata such as time, date, and GPS coordinates.
- Converts audio captured into a decibel value and uploads to the cloud server.

6.1.2.2 Pre-processing and Feature Extraction Module

- Preprocesses raw audio input and extracts meaningful acoustic features such as MFCCs, spectral centroid, and zero-crossing rate using libraries such as Librosa.
- Formats and normalizes data for classification.

6.1.2.3 Noise Classification Module

- Uses supervised machine-learning models (SVM, CNN, or Random Forest) to classify sources of sound into a predefined list of categories such as light traffic, heavy traffic, human speech, construction, or background noise.

6.1.2.4 Visualization and User Interface Module

- Provides an interactive web dashboard built using mapping libraries (e.g., Leaflet.js or Mapbox).
- Displays noise intensity as color-coded heatmap.
- Includes a time-lapse mode for day/night and temporal visualisation.
- Presents data insights through charts, reports, and summary distribution of noise-type.

6.2 Design Considerations

This section outlines the critical design considerations, assumptions, dependencies, constraints, and guiding principles that influenced the design of the EchoMap system.

6.2.1 Assumptions and Dependencies

Assumptions or dependencies regarding the software and its use:

6.2.1.1 Related software or hardware :

- The system assumes customers own smartphones with operational microphones and GPS modules as noise collection devices.
- The internet connection must be stable (Wi-Fi or mobile network) to transfer data to the cloud server.
- Frontend development is built on top of web technologies (HTML, CSS, JavaScript, and mapping libraries such as Leaflet.js or Mapbox).
- Back-end is built on Python-based frameworks (Flask/Django) and machine learning libraries (TensorFlow, Scikit-learn, Librosa).
- Persistent data storage will require support of cloud databases (Firebase, MongoDB, or SQL).

6.2.1.2 Operating systems :

The web application will be cross-platform, compatible with recent browsers on Windows, Android, iOS, and macOS

6.2.1.3 End-user characteristics :

- Users need to be digitally literate in general in order to access and use the web dashboard and mobile interface.

- Users of the analytics tool, for example city managers or environmental researchers, are assumed to be already acquainted with data visualization.

6.2.1.4 possible and/or likely changes in functionality :

The design is scalable in the form of anticipating future growth through inclusion of more sensors, other cities, or more upscale features such as real-time notice and predictive modeling.

6.2.2 General Constraints

Below are constraints that have significant impacts on the EchoMap system design and implementation:

6.2.2.1 Hardware and Software Environment :

Since smartphone microphones are not uniformly sensitive, measurements in decibels would have minor calibration errors. The software needs to be light enough to execute well on mid-range phones.

6.2.2.2 Network and Resource Availability :

- Design is largely dependent upon internet connectivity to transfer data. Offline functionality is not a major design goal.
- Cloud server storage space and bandwidth constraints restrict the size of real-time uploads.

6.2.2.3 Performance and Capacity:

- The backend should be able to handle multiple concurrent data uploads without delay.
- Data lookup in low-latency for heatmap visualization should be supported by indexing and caching strategies in the database.

6.2.2.4 Security and Privacy Constraints:

- User authentication and data encryption are needed in order to protect sensitive information (location, timestamps).
- Personal information must comply with data protection regulations (such as GDPR standards).

6.2.2.5 Interface/protocol requirements :

The mapping and visualization component should support open-source APIs and web protocol standards (HTTP/HTTPS, REST APIs).

6.2.2.6 Verification and Validation Requirements:

All the modules—data collection, ML classification, and visualization—must undergo unit testing and integration testing to provide reliable performance.

6.2.3 Goals and Guidelines

The EchoMap design adheres to some guiding principles to maintain usability, performance, and ease of maintenance.

The KISS principle ("Keep it simple stupid!") : The architecture is designed with modular components to facilitate easy maintenance and future extensions. Each module (data collection, ML calculation, visualization) can be independently modified.

Scalability and Extensibility : The design allows for future growth, e.g., integration of real-time IoT sensors or multi-city implementation, without a lot of redesigning.

User-Centric Design: The GUI is focused on simplicity and usability, presenting a natural user experience to end users and pros alike through clean design and dynamic layouts.

6.2.3.1 Efficiency and Performance Optimization:

- Resource-efficient management becomes the priority by optimizing data maximally, using caching, and minimal front-end rendering to provide responsiveness even under heavy data loads.
- ML algorithms are trained using various datasets to make noise sources predictable and reliable classification, and noise intensity correct visualization.

6.2.4 Development Methods

EchoMap development is performed with an iterative and modular software development process, primarily based on the Agile methodology.

Agile Development: The software is developed in iterative sprints with ongoing testing, user input, and feature increment.

Incremental Design: The core building blocks—data acquisition, ML prediction, visualization—are incrementally built, with each feature validated early in the process.

Prototyping Approach: Low-fidelity drawings and Figma GUI mockups are created to test user interface designs before implementation.

Toolchain and Frameworks: Python (backend and ML), JavaScript (frontend mapping and visualiza-

tion), and cloud platforms (deployment) form the basis of development.

This approach assures flexibility, rapid development, and frictionless interaction between frontend, backend, and data processing teams.

6.3 System Architecture

The system architecture of EchoMap is designed as a modular and scalable three-tier framework that integrates mobile data collection, machine learning-based processing, and web visualization into one system. The project has its primary goal to make it easier for the users to record environmental noise by means of their smartphones and, thereafter, to transform raw data into valuable insights provided through an interactive map interface. The last or client layer, which is described as the outer layer, is where the mobile and web user interface is located, as well as the places where users can sample sounds, show heatmaps, and read analytical reports. User interfaces have been developed using technologies such as Leaflet.js or Mapbox for map rendering and are intended to provide a very pleasant and low-latency user experience. The second layer, i.e., application or backend layer, is responsible for all the logic and processing of data. It receives the data from the client, extracts valuable features from the audio files with the help of libraries like Librosa, and distinguishes the types of noise using supervised machine learning techniques such as SVM, CNN, or Random Forest. This layer provides seamless interaction between data ingestion, processing, and visualization modules. The third level is the data management level, which accumulates all gathered and processed data into a structured system database (NoSQL or SQL). It holds user records, noise value intensities, classification outcomes, and spatial data for mapping. Optional integration with external systems like the Pakistan Environmental Protection Agency API for bringing real-time or historical noise data is also supported in the architecture. The resulting data is then represented by means of an interactive dashboard that offers capabilities like heatmaps, time-lapse playback, and granular statistical reports. Modularity of the system makes each module updateable or scalable independently in order to provide flexibility for future extensions like city-wide deployment or real-time tracking. This architecture in total prioritizes efficiency, usability, and extensibility while facilitating significant data-driven insights for citizens as well as policy makers.

6.3.1 Subsystem Architecture

The subsystem architecture of EchoMap divides the entire system into smaller, manageable pieces that work cohesively towards the ultimate goals of the project. The system is composed of five primary subsystems: Data Collection, Data Processing, Machine Learning, Visualization, and Database Management. The Data Collection Subsystem functions through the mobile application, which captures the

ambient noise level, GPS coordinates, and timestamps, sending this information in structured format to the backend. The Data Processing Subsystem takes this input, does the preprocessing like normalization, feature extraction, and filtering utilizing libraries such as Librosa, and sets up the data for classification. The Machine Learning Subsystem then applies pre-trained SVM, CNN, or Random Forest classifiers to classify the noise into classes like human voice, light traffic, or heavy traffic. The Database Management Subsystem is the core unit for storing raw audio samples, extracted features, classification outputs, and user profiles in a secure and query-efficient way. It also combines data from external APIs such as the Pakistan EPA to enhance the dataset. Lastly, the Visualization Subsystem processes stored data into comprehensible formats like heatmaps and graphical reports using Leaflet.js or Mapbox, and temporal visualization to display day-night differences in noise pollution. All subsystems interact through RESTful APIs for seamless and modular exchange of data. This layered design not only improves system reliability and maintainability but also promotes scalability by enabling a subsystem to be upgraded or replaced without impacting others. Through the organization of the system into well-defined functional units, the subsystem architecture facilitates efficiency, transparency, and versatility, which are essential for the sustainable development of an intelligent environmental monitoring platform such as EchoMap.

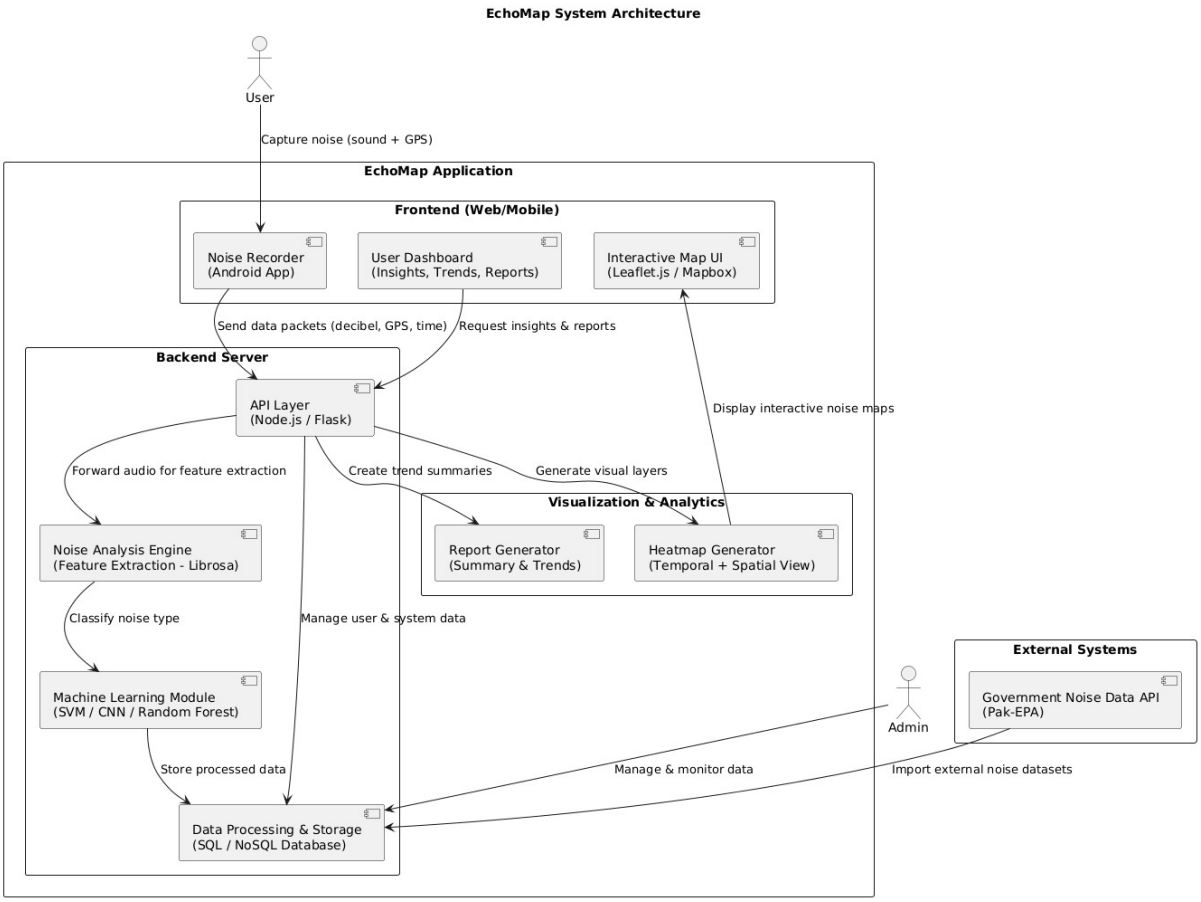

Figure 6.1: Architecture Diagram

6.4 Architectural Strategies

The main architectural methods and design choices, which were applied in the formulation of EchoMap, the intelligent noise pollution mapping and analysis system, are discussed in this section. The said methods have determined the basic composition, the main abstractions and the operations of the system. Every single method reflects a balancing act between scalability, maintainability, performance, and usability.

6.4.1 Layered and Modular Architecture

Description: The EchoMap system employs a three-layer arrangement consisting of the Presentation Layer (frontend), Application Layer (backend API), and Data Layer (database + ML models).

Reasoning: The modular structure facilitates maintenance, allows for simultaneous development, and makes it possible to scale the system. The communication between the layers occurs through properly defined RESTful APIs which lead to loose coupling.

Trade-offs:

- **Advantages:** Improved structuring of code, easier debugging, independent scaling.
- **Disadvantages:** Slightly increased latency because of communication between more layers.
- **Alternative contemplated:** Monolithic architecture — abandoned due to poor scalability and maintainability.

6.4.2 RESTful Communication Mechanism

Description: The entire communication between the frontend and backend happens through the RESTful APIs that are created using the Django REST Framework (DRF).

Reasoning: RESTful APIs enable a stateless and standard communication model, which thus grants interoperability among web and mobile clients.

Trade-offs:

- **Benefits:** Platform independence, simplicity, and popularity.
- **Drawbacks:**Overhead involved in redundant data requests.
- **Alternative:** WebSocket for live updates — to be utilized in the future when streaming of live data is required.

6.4.3 Machine Learning Pipeline Integration

Description: The machine learning module which is built with the help of Python and based on the libraries like Librosa, scikit-learn, and TensorFlow is completely a separate unit from the whole system. It is responsible for extracting features from audio, training, and predicting for noise classification.

Reasoning: By taking the ML reasoning apart from the main application, it becomes much easier to update the models in the future with a very small impact on the rest of the system.

Trade-offs:

- **Advantage:** Easy retraining and swapping in improved models.
- **Disadvantages:** Additional CPU and storage requirements.
- **Alternative:** Hard-coded rule-based classification — eliminated due to low accuracy and inflexibility.

6.4.4 Geospatial Data Management

Description: Utilize PostgreSQL with PostGIS extension for spatial data storage and management such as GPS coordinates and noise level intensity.

Reasoning: PostGIS allows for performing advanced spatial queries (for example, finding the noise hotspots within a specified radius) and map rendering with great efficiency.

Trade-offs:

- **Advantages:** Accurate spatial analysis, query optimization.
- **Disadvantages:** Relatively complex database setup and higher learning curve.
- **Alternative:** NoSQL (MongoDB with GeoJSON)

6.4.5 Data Visualization and UI Paradigm

Description: The frontend uses a combination of React.js and Leaflet.js to produce noise heatmaps that are dynamic and interactive, as well as filters and time-lapse playback.

Reasoning: React ensures UI management via components, whereas Leaflet makes it possible to render maps together with the overlay of geo-data in an efficient manner. The system is capable of delivering both day and night functionalities along with a mobile-friendly responsive design.

Trade-offs:

- **Pros:** Smooth user experience, support for large datasets.

- **Cons:** Slightly elevated rendering burden on low-end devices.
- **Alternative:** Static map images — fell through due to inability to interact with them.

6.4.6 Error Handling and Recovery

Description: Implement centralized error handling in the backend through Django middleware and error logging. The frontend notifies users with pleasant messages in case of network or data problems.

Reasoning: Reliability, traceability, and better debugging are ensured. Logs are stored for later analysis and maintenance.

Trade-offs:

- **Advantages:** Increased fault tolerance and maintainability.
- **Disadvantages:** Slightly higher logging overhead.
- **Alternative:** Distributed manual error catching — declined as error-prone and unreliable.

6.4.7 Concurrency and Performance Optimization

Description: Django's asynchronous request handling (ASGI support) and frequent API response (e.g., map data) caching using Redis will be employed.

Reasoning: Concurrent user loads will be handled more efficiently and server response times for static or repeated queries will be shortened.

Trade-offs:

- **Pro:** Great scalability, lower latency.
- **Con:** Ever so slightly more complicated deployment setup.
- **Historic synchronous processing** — abandoned due to inefficiency under multiple concurrent users.

6.4.8 Future Extensibility

Description: The structure of the design is provided for scaling and modular extension. Future versions can incorporate:

- Noise source prediction based on AI, and
- Integration of municipal APIs for environmental sensing.

6.5 Domain Model/Class Diagram

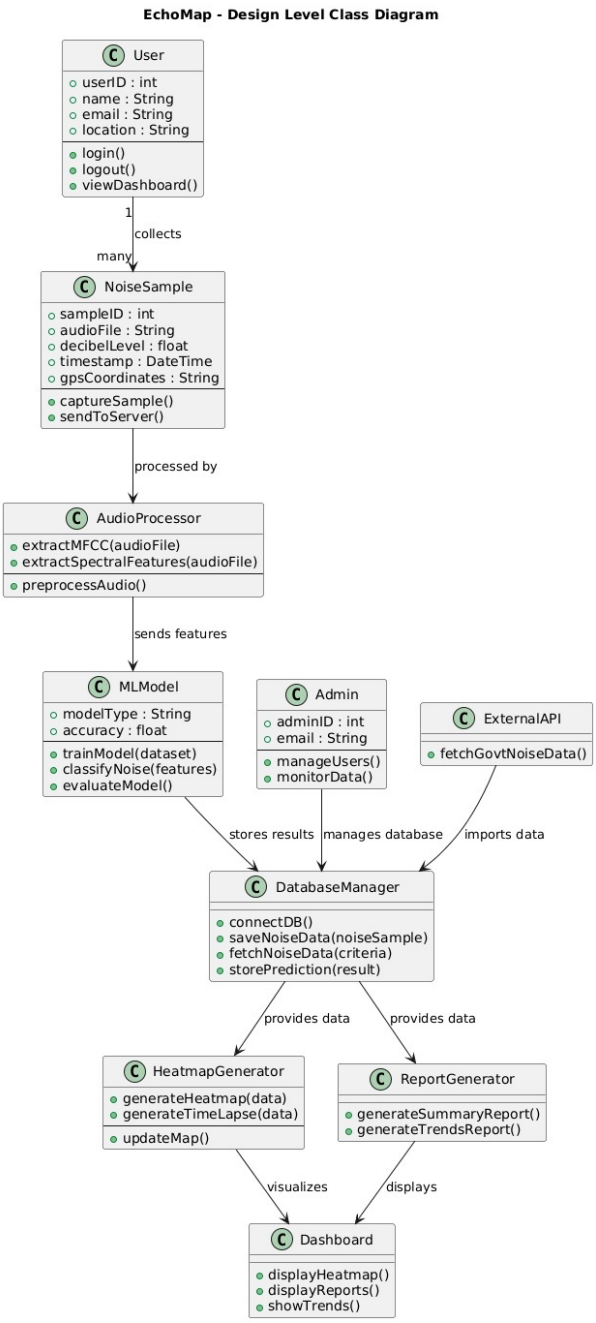

Figure 6.2: Class Diagram

6.6 Policies and Tactics

This section describes design policies and implementation strategies for EchoMap, which influence coding conventions, tool choice, testing, and maintainability. They don't have any impact on the system design but ensure consistent implementation quality, performance, and maintainability.

6.6.1 Technology Stack and Tool Selection

6.6.1.1 Decision:

Use Python as the backend platform for machine learning and data processing, Django REST Framework (DRF) for API building, and React + Leaflet.js for web visualization interface. Database: PostgreSQL with PostGIS extension to support geospatial data.

6.6.1.2 Rationale:

- Python provides mature ML libraries (Librosa, scikit-learn, TensorFlow).
- Django REST enables fast API development and secure authentication.
- Leaflet.js / React support responsive, interactive map visualizations.
- PostGIS supports efficient spatial queries for noise-level mapping.

Alternative considered: Node.js + Express; rejected due to it having fewer ML library resources than Python.

6.6.2 Coding Guidelines and Conventions

6.6.2.1 Decision:

Follow PEP 8 for Python and ESLint + Prettier for JavaScript/React. Code is organized into logical units: data gathering, model training, API services, frontend views, and utilities.

6.6.2.2 Rationale:

Consistent style improves readability and collaboration productivity. Naming convention and docstrings ensure traceability and maintainability.

6.6.3 Machine Learning Algorithm Selection

6.6.3.1 Decision:

Use MFCC and spectral features for feature extraction, and Random Forest for standard classification and CNN for future work.

6.6.3.2 Rationale:

- Random Forest provides interpretability and fast training on small data.
- CNNs offer better accuracy for complex acoustic patterns.
- This pairing provides a trade-off between accuracy and efficiency.

Alternative under consideration: Support Vector Machine (SVM); ruled out due to longer training time and lower scalability for real-time inference.

6.6.4 Testing and Validation

6.6.4.1 Decision:

Create unit test for backend functions (PyTest), API test using Postman, and front-end test using Jest. Test ML models using cross-validation.

6.6.4.2 Rationale:

Ensures correctness of individual modules, API endpoints, and UI elements. High reliability as the system increases in size.

6.6.5 Requirements Traceability and Documentation

6.6.5.1 Decision:

Maintain traceability matrix linking requirements, design elements, and test cases through GitHub Projects and Markdown documentation.

6.6.5.2 Rationale:

Facilitates transparency and consistency between implementation, design, and requirements. Supports version tracking and future changes.

6.6.6 Software Maintenance and Deployment

6.6.6.1 Decision:

Use Docker for containerization and GitHub Actions for continuous integration (CI). Scheduled model retraining will be facilitated through scheduled scripts.

6.6.6.2 Rationale:

- Docker supports platform independence and reproducibility.
- CI/CD automatically validates and deploys with no room for human error.
- Retraining ensures that the model is accurate using new noise data.

6.6.7 User Interface and Usability Design

6.6.7.1 Decision:

- Deploy mobile-first responsive design with light/dark (day/night) themes to enhance accessibility.
- Employ user-friendly color-coded heatmaps with legends and filters for date/time.

6.6.7.2 Reasoning:

- Enhances user experience for mobile and desktop users.
- Enables environmental authorities and citizens to quickly identify noise hotspots.

6.7 Conclusion

The High-Level and Low-Level Design of EchoMap serves as a blueprint for the system, outlining the overall architecture, major tactics, and the policies for implementation that prioritize scalability, support, and performance. Furthermore, it implements a layered and modular architecture with a distinct separation of the presentation, application, and data layers, which are connected via RESTful APIs created with the Django REST Framework. Additionally, the system includes a machine learning pipeline (Librosa, scikit-learn, TensorFlow) for the classification of noises and operates with PostgreSQL with PostGIS for the management of geospatial data that is very fast and efficient. The user interface utilizes 'React.js' and 'Leaflet.js' to create interactive heatmaps and visualizations, whereby backing operations are also asynchronous and caching is done for performance. The design policies put a premium on the quality of the code, and adherence to consistent conventions (PEP 8, ESLint), powerful testing and validation, and continuous integration/deployment through Docker and GitHub Actions. Anticipated future

expansion involves the real-time integration of IoT and the application of AI in noise prediction. In general, the design has the attributes of being scalable, fast, and user-friendly, thus, the system is designed for monitoring urban noise and analyzing it effectively.

Chapter 7 Implementation and Test Cases

In FYP-1, we developed the initial prototype components of EchoMap to illustrate the core idea: real-time noise monitoring, mapping, and classification. Although the system is not fully developed yet, several foundation modules have been implemented in order to validate its feasibility and present the workflow during the mid-term evaluation. Currently, the prototype includes a basic data-collection interface, some initial backend endpoints, sample visualizations, and a preliminary machine-learning processing pipeline. These form the building blocks for the full system that will be developed in FYP-2.

7.1 Implementation

The technical implementation for each component of the prototype is explained in the subsections that follow.

7.1.1 Noise Data Collection Prototype

To simulate the actual data-collection process, a simple web-based module has been developed that not only captures audio samples using the device microphone but also the current GPS location of the user. Recording: Audio recording is done via Web Audio API. When the noise level is estimated, a decibel approximation function depends on amplitude values. Each recording is tagged with latitude, longitude, and timestamp extracted using the browser's Geolocation API. The sample is then sent, in JSON format, to the backend for processing. This allows us to test data flow, structure, and communication between the client and server before developing the full-scale mobile interface.

7.1.2 Backend API and Data Pipeline

A simple backend service using Python Flask was designed that provided the functionality of receiving audio clips, calculating initial thresholds, and storing data in a temporary database for analysis. The backend currently handles:

- Upload audio samples
- Audio preprocessing - Normalization, trimming, conversion to .wav
- Save all metadata: noise level, location, time

This pipeline will be further expanded into a complete machine learning processing module in FYP-2.

7.1.3 Audio Feature Extraction Module

A prototype feature extractor was implemented in Python to test whether our system can successfully process different types of noise. The extractor calculates:

- MFCCs
- Zero-crossing rate
- Spectral centroid
- Chroma features

This module ensures that typical smartphone recordings yield extractable audio features that could be useful in machine learning classification. This prototype currently processes offline audio samples that will be replaced later with live data from the frontend.

7.1.4 Initial Noise Classification Model

- We have trained a small test model using a subset of all publicly available urban-sound datasets to evaluate classification feasibility.
- The prototype uses a simple Random Forest and SVM model that classify sample noise clips into categories such as human voice, traffic or background noise.
- The goal of this step was not to achieve perfect accuracy but rather to check that the chosen features and formats of datasets are compatible with the machine-learning pipeline.

The model will be upgraded to CNN-based architectures in FYP-2 for better accuracy.

7.1.5 Original Interactive Map Prototype

Below is an initial map visualization built with Leaflet.js, showing how noise readings will be plotted geographically. Implemented features include:

- Display user location
- Setting sample markers that represent noise levels Color-coded circles indicate low, medium, and high intensity of noise. Interactive popup display showing the decibel value, time, and predicted noise type This prototype confirmed that mapping, placing pins, and heat representation are fully compatible with the libraries we chose

7.1.6 Heatmap Day/Night View Simulation

An additional, basic heatmap layer was added to test visual density representation. The heatmap uses sample data points to simulate noise hotspots. Two different layers showing day and night view, which can be toggled for the demonstration of the UI concept. Now the system is showing mock values, but the structure is This will be complete and connected to live data in FYP-2.

7.1.7 Dashboard UI Prototype

With ReactJS, a simple front-end dashboard was designed in order to preview how users will interact. with the system. The dashboard includes:

- A section for data upload/recordings
- A map visualization page
- Basic statistics: average noise, highest recorded noise, most common noise types
- A navigation layout for reports and analysis. Although still at a minimum, this interface demonstrates the layout and overall visual direction for the final product.

7.2 Conclusion:

The implementation phase of FYP-1 has thus laid the foundation for EchoMap, which is now A functional prototype, bringing the initial concept to life: We demonstrated the feasibility of the proposed workflow from a technical point of view by developing and testing the needed components like the web-based data collection interface, Flask backend pipeline, audio feature extraction modules, Interactive mapping visualizations - currently, simple machine learning is already applied within the present system. models using Random Forest and SVM with simulated data streams, but these test cases have shown that the selected libraries and data structures are indeed good. This technical base, quite Sturdy has been set for the next phase of the project FYP-2, in which the development will be shifted to Convolutional Neural Networks, or CNNs, will be combined with the capability of processing data in. real-time and completion of the full-scale mobile application.

Chapter 8 Experimental Results and Discussion

Introduction

This chapter describes the experimental work carried out during the development of EchoMap. The core The idea behind the experiment was to understand how noise pollution data can be collected, processed, visualized with affordable and easy-to-access technologies. We reviewed various tools and frameworks, tested the mobile-based decibel measurement apps, and tried to access real-world noise datasets. The aim was to validate the feasibility of our proposed system and highlight practical challenges while implementing a smart noise mapping solution. Our focus was on understanding how smartphone microphones perform in real-world conditions, how Noise types can be classified using Machine Learning accurately, and how effective the data can be visualized on a map. We also wanted to see how users interact with the system and what kind of insights they find useful.

8.1 Experimental Procedure

Ambient recordings were made, as a starting point, in a series of small-scale experiments using mobile phones. Noise in different locations like busy roads, university corridors, parks, and residential areas. The recorded audio samples were further processed using Librosa for feature extraction like MFCC and spectral features. Contrast. We also built a basic classification model using scikit-learn and tested it on labeled samples to identify noise types like traffic, human voices, and construction sounds. For visualization, we used Leaflet.js to plot the noise levels on a map and added a simple time-lapse feature that showed variations during day and night. The following were the major areas of experimentation:

- Testing smartphone microphone accuracy for decibel measurement.
- Collecting GPS-tagged audio samples from different environments.
- Audio feature extraction and training of ML models for classification.
- Noise data visualization with heatmaps and time-lapse playback.
- Assessing the interaction of users with the dashboard prototype.

8.2 Results and Analysis

8.2.1 Smartphone Microphone Performance

We have found that though smartphone microphones are not as accurate as professional sensors, they can nonetheless provide reasonably consistent decibel readings in controlled conditions. However, background noise and positioning of the device therefore affected the readings, meaning that calibration is necessary for better accuracy.

8.2.2 Noise Classification Accuracy

Our first ML model, using MFCC and spectral features, achieved about 78 percent accuracy in classifying types of noise. Traffic noise was the most easily identified, whereas overlapping sounds-like voices mixed with Traffic was even more difficult to categorize. We expect to enhance this using more sophisticated models such as CNNs in the next phase.

8.2.3 Heatmap Visualization

The interactive map succeeded in mapping out the noise intensity across locations. Users could easily Identify hotspots and observe how noise levels changed over time. The time-lapse feature helped to allow for envisioning the pattern of increased traffic noise during rush hours and quieter periods at night.

8.2.4 User Feedback

We shared the dashboard with a few students and faculty members for feedback. Most users appreciated They enjoyed visual representation and found it useful regarding noise pollution in general.

8.3 Summary of Experimental Results

Observation Category	Result	Interpretation
Smartphone microphone accuracy	Moderate	Needs calibration for reliable decibel readings
Noise classification accuracy	78%	Promising results with scope for improvement using deep learning
Heatmap usability	High	Users found it intuitive and informative
Time-lapse feature	Useful	Helped visualize daily noise patterns
User feedback	Positive	Suggested enhancements for filtering and customization

Table 8.1: Summary of experimental observations

8.4 Discussion

The experiments done during FYP -I helped us understand the practical aspects of building a noise mapping system. Though smartphone-based data collection is possible, it would have to be calibrated and preprocessing to ensure accuracy. Results from the classification showed that machine learning can be effectively used to identify noise types, though more training data and better models are required.

Two most valuable means for communication were visualization through heatmaps and time-lapse playback. noise trends. It enables the user to meaningfully explore the data and supports decision-making for urban planning and awareness.

Overall, the experimental phase validated our concept by showing areas that needed further improvement. Besides that, It gave us the confidence that indeed, EchoMap has the potential to become a useful noise-pollution monitoring platform.

8.5 Conclusion

This chapter summarized the experimental work done in FYP-I, including data collection, classification,and visualization. We tested the feasibility of using mobile devices and open-source tools to build a Smart Noise Mapping System: the results were encouraging, and provided a sound basis for further development in FYP-II. With better models, superior calibration, and additional user features, EchoMap can be turned into a valuable tool in environmental monitoring and smart city planning.

Chapter 9 Conclusions

Work done in this phase of the project thus represents an important milestone towards the development of a functional, intelligent noise-pollution monitoring and visualization system. In the FYP-1, we have investigated the deficiencies of current approaches to assess urban noise and analyzed state-of-the-art methods for coupling mobile sensing, geospatial mapping, and machine learning. This allowed us to determine with clarity the deficiencies of current practices in noise monitoring: almost a lack of low-cost solutions, real-time insights, and contextual understanding of noise sources. Our study confirmed that a mobile-based, machine-learning-supported platform such as EchoMap can address such deficiencies through more accessible and data-driven analysis for both citizens and planners.

The system goals were settled in this phase, the technical requirements outlined, a broad literature review performed, and the main tools to be used in the system analyzed: Librosa, scikit-learn, TensorFlow, and Leaflet.js. We have also explored how decibel capturing has been done in relation to mobile platforms, made an audio classification approach, and prepared the architectural flow of the complete system. Therefore, we confirmed the project's viability, setting a very clear basis for further development. Even though the system has not been implemented yet, all major design decisions were drafted along with methodologies and workflows to enable the next phase without problems.

The scope set for FYP-1 was fully covered. We did the research, finalized the functional boundaries, and decided which features are achievable with the available time, data, and equipment. The objectives set for the project at this stage, which include planning a low-cost monitoring solution, identifying appropriate machine learning techniques, designing the mapping mechanism, and formulating the time-lapse visualization strategy, have been achieved. However, access to noise datasets provided by the government is still pending, and thus the calibration study was slightly delayed. This did not affect the overall progress but will require follow-up in the implementation phase. Large-scale deployment and integration into official policy were excluded intentionally since they are beyond the approved scope and need long-term institutional collaboration.

It has been demonstrated that, ultimately, FYP-1 attained its objective in creating a sound conceptual basis, reviewing necessary enabling technologies, and drafting a roadmap for the actual construction of the EchoMap system.

9.1 Future Work (FYP-2 Plan)

FYP-2 work flow will be:

9.1.1 Data Collection Pre-processing

- Collect audio samples from smartphones and available EPA datasets.
- Audio clip labeling, noise cleaning, and segmentation.

9.1.1.1 Machine Learning Model Development

- Training and Comparison of Machine Learning Algorithms, SVM, CNN, Random Forest, on Noise Type Classification.
- Test the accuracy, precision, and generalization performance.
- Optimize the model for mobile-based or server-based inference.

9.1.1.2 Backend and API Development

- Data processing pipeline will be build that receives audio, extracts features, performs inference, and saves the results.
- Create APIs for the visualization modules and the user's dashboards.

9.1.1.3 Mapping Visualization Module

- Create an interactive heat map using either Leaflet.js or Mapbox.
- Combine the spatial and temporal filters.
- Construct the day/night time-lapse noise playback feature.

9.1.1.4 Dashboard Frontend Development

- Provide an intuitive interface to display noise level, trend, and hotspots.
- Add analytics such as hourly averages, peak noise alerts, and classification summaries. Pilot testing in a selected region, for example, the FAST campus or adjacent sectors.

9.1.1.5 Testing, Evaluation Refinement

- Pilot testing in a selected region, such as the FAST campus or adjacent sectors.
- Compare the predicted noise labels against real-world ground truth Development details, results, model comparisons, and system evaluation should be compiled.

9.1.1.6 Documentation Final Report

- Provide deployment instructions and suggest areas of future research.
- Mobile version of the application with real-time alerts.
- Noise reporting crowdsourced with game-style incentives.
- Expanding into a multi-city deployment with cloud-based storage and analytics.

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